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# GEOGRAPHY

## GRADE 9 · ጂኦግራፊ

## Unit 1: Geological History and Topography of Ethiopia

## ምዕራፍ 1፦ የኢትዮጵያ የጂኦሎጂ ታሪክና የመሬት አቀማመጥ

EUEE PREPARATION EDITION · BILINGUAL

English · አማርኛ

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# How to Use This Book

## ይህን መጽሐፍ እንዴት መጠቀም እንደሚቻል

### ENGLISH

This book turns the Grade 9 Geography Unit 1 notes into a complete, exam-ready study guide. Every idea is written in English and in Amharic, so you can learn in the language you understand best and still know the English words the exam uses.

### Recommended study plan (10 days)

- Days 1–2: Sections 1.1 (meaning, scope, branches). Learn the definition and the two branch trees.
- Days 3–4: Section 1.2 (location, size, shape). Practise the three location types and both compactness calculations.
- Days 5–8: Section 1.3 (geological history and landforms). Learn the eras as a timeline, then the three physiographic divisions.
- Day 9: Summary tables and Exam Strategy (mnemonics). Say the tables aloud from memory.
- Day 10: Take the 40-question bank under time (45 minutes), mark it with the answer key, and re-read the sections behind every mistake.

### Level tags in the question bank

- **Easy** – direct recall of a fact from the notes.
- **Medium** – needs understanding, comparison or a short calculation.
- **Hard** and **EUEE Trap** – several ideas at once, or a wrong option that looks right. Read every option before choosing.

### Boxes you will meet

- **Worked Example** (gold) – a solved problem or model answer.
- **Common Misconceptions** (red) – mistakes students make and the correct fact.
- **Key Takeaways** (blue) – the points to memorise from each major topic.

### አማርኛ

> ይህ መጽሐፍ የ9ኛ ክፍል ጂኦግራፊ ምዕራፍ 1 ማስታወሻዎችን ወደ ሙሉ የፈተና ዝግጅት መመሪያ ይቀይራል። እያንዳንዱ ሐሳብ በእንግሊዝኛና በአማርኛ ተጽፏል፤ ስለዚህ በሚገባዎት ቋንቋ ተምረው ፈተናው የሚጠቀምባቸውን የእንግሊዝኛ ቃላት ያውቃሉ።

### የሚመከር የጥናት ዕቅድ (10 ቀናት)

- ቀን 1–2፦ ክፍል 1.1 (ትርጉም፣ ወሰን፣ ቅርንጫፎች)። ትርጓሜውንና ሁለቱን የቅርንጫፍ ዛፎች ይማሩ።
- ቀን 3–4፦ ክፍል 1.2 (አቀማመጥ፣ ስፋት፣ ቅርጽ)። ሦስቱን የአቀማመጥ ዓይነቶችና ሁለቱን የውሱንነት (compactness) ስሌቶች ይለማመዱ።
- ቀን 5–8፦ ክፍል 1.3 (የጂኦሎጂ ታሪክና የመሬት ቅርጾች)። መጀመሪያ ዘመናትን እንደ የጊዜ መስመር ይማሩ፤ ከዚያ ሦስቱን የመሬት አቀማመጥ ክፍሎች።
- ቀን 9፦ የማጠቃለያ ሰንጠረዦችና የፈተና ስልት (ማስታወሻ ዘዴዎች)። ሰንጠረዦቹን በቃል ይናገሩ።
- ቀን 10፦ የ40 ጥያቄዎችን ስብስብ በጊዜ (45 ደቂቃ) ይስሩ፤ በመልስ ቁልፉ ያርሙ፤ ለተሳሳቱባቸው ጥያቄዎች ተጓዳኝ ክፍሎችን ደግመው ያንብቡ።

### በጥያቄ ስብስብ ውስጥ ያሉ የደረጃ ምልክቶች

- **Easy (ቀላል)** – ከማስታወሻው ቀጥተኛ እውነታ ማስታወስ።
- **Medium (መካከለኛ)** – መረዳት፣ ማወዳደር ወይም አጭር ስሌት የሚጠይቅ።
- **Hard (ከባድ) እና EUEE Trap (የፈተና ወጥመድ)** – ብዙ ሐሳቦችን አንድ ላይ የሚጠይቅ ወይም ትክክል የሚመስል የተሳሳተ አማራጭ ያለው። ከመምረጥዎ በፊት ሁሉንም አማራጮች ያንብቡ።

### የሚያጋጥሙዎት ሳጥኖች

- **Worked Example (የተሰራ ምሳሌ)** (ወርቃማ) – የተፈታ ጥያቄ ወይም ናሙና መልስ።
- **Common Misconceptions (የተለመዱ የተሳሳቱ ግንዛቤዎች)** (ቀይ) – ተማሪዎች የሚሳሳቱባቸውና ትክክለኛ እውነታዎች።
- **Key Takeaways (ዋና ነጥቦች)** (ሰማያዊ) – ከእያንዳንዱ ዋና ርዕስ በቃል የሚያዙ ነጥቦች።

**HONEST NOTE ABOUT SOURCES · ስለ ምንጮች ሐቀኛ ማሳሰቢያ**

All facts come from the Unit 1 study notes (StudySmart G9 Geography, Unit 1). The 40 practice questions are **original EUEE-style questions** written for this book; they are not copies of real past papers, and no past-paper source was available to verify against.

ሁሉም እውነታዎች ከምዕራፍ 1 የጥናት ማስታወሻ የተወሰዱ ናቸው። 40ዎቹ የመለማመጃ ጥያቄዎች ለዚህ መጽሐፍ የተጻፉ ኦሪጅናል የ**EUEE** አይነት ጥያቄዎች ናቸው፤ ከእውነተኛ የቀድሞ ፈተናዎች የተገለበጡ አይደሉም።

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# Unit Introduction

## የምዕራፉ መግቢያ

### ENGLISH

Unit 1 explores the foundations of Geography: what it is, what it studies, and how it is divided. It then moves to Ethiopia itself, asking where the country lies, how large it is and what its shape means for defence and administration. Finally it follows the geological history of Ethiopia across four great eras to explain why the country has plateaus, mountains, gorges, lowlands and a Rift Valley.

### The unit in one picture

- **1.1** Geography: meaning, scope and branches.
- **1.2** Location, size and shape of Ethiopia.
- **1.3** Geological history and landforms of Ethiopia.

### Why it matters for the exam

- Definitions and branch names are easy marks if you learn the exact wording.
- Numbers (3°N–15°N, 1,106,000 km², 5,260 km, 4,620 m, −116 m, 56 %, 44 %, 18 %) appear again and again.
- Chronology questions (which era or period formed which rock or landform) are the most common source of mistakes.

### አማርኛ

> ምዕራፍ 1 የጂኦግራፊን መሠረቶች ይቃኛል፦ ምንድን ነው፣ ምን ያጠናል፣ እንዴትስ ይከፋፈላል። ከዚያም ወደ ኢትዮጵያ ይዘረልፏ፤ አገሪቱ የት እንደምትገኝ፣ ምን ያህል ስፋት እንዳላት፣ ቅርጿም ለመከላከያና ለአስተዳደር ምን ማለት እንደሆነ ይመረምራል። በመጨረሻም ኢትዮጵያ ለምን ሰፊ አምባዎች፣ ተራሮች፣ ጥልቅ ሸለቆዎች፣ ዝቅተኛ ሜዳዎችና ስምጥ ሸለቆ እንዳላት ለማስረዳት የጂኦሎጂ ታሪኳን በአራት ታላላቅ ዘመናት ይከተላል።

### ምዕራፉ በአንድ እይታ

- **1.1** ጂኦግራፊ፦ ትርጉም፣ ወሰንና ቅርንጫፎች።
- **1.2** የኢትዮጵያ አቀማመጥ፣ ስፋትና ቅርጽ።
- **1.3** የኢትዮጵያ የጂኦሎጂ ታሪክና የመሬት ቅርጾች።

### ለፈተናው ለምን ወሳኝ ነው

- ትርጓሜዎችና የቅርንጫፍ ስሞች ትክክለኛውን አገላለጽ ከተማሩ ቀላል ነጥቦች ናቸው።
- ቁጥሮች (3°N–15°N፣ 1,106,000 ካ.ኪ.ሜ፣ 5,260 ኪ.ሜ፣ 4,620 ሜ፣ −116 ሜ፣ 56 %፣ 44 %፣ 18 %) ደጋግመው ይመጣሉ።
- የዘመን ጥያቄዎች (የትኛው ዘመን ወይም ክፍለ ዘመን የትኛውን ዐለት ወይም የመሬት ቅርጽ ፈጠረ) በብዛት የስህተት ምንጭ ናቸው።

# Learning Objectives

## የትምህርቱ ዓላማዎች

### ENGLISH

After completing this unit you will be able to:

አማርኛ

> ይህን ምዕራፍ ከጨረሱ በኋላ የሚከተሉትን ማድረግ ይችላሉ፦

1. Define geography and explain where the word comes from.
> ጂኦግራፊን መተርጎምና የቃሉን ምንጭ ማብራራት።

2. Name Eratosthenes, Humboldt, Hartshorne and Yeates and state each one's idea.
> ኤራቶስቴንስን፣ ሁምቦልትን፣ ሃርትሾርንንና የትስን ስሞችና ሐሳባቸውን መግለጽ።

3. Describe the scope of geography, including the five spheres.
> የጂኦግራፊን ወሰን ከአምስቱ ስፌሮች ጋር መግለጽ።

4. Identify the two main branches of geography and their sub-branches.
> ሁለቱን ዋና የጂኦግራፊ ቅርንጫፎችና ንዑስ ቅርንጫፎቻቸውን መለየት።

5. Distinguish relative (vicinal and strategic) from absolute location.
> አንጻራዊ (አጎራባች እና ስትራቴጂያዊ) አቀማመጥን ከፍጹም አቀማመጥ መለየት።

6. State Ethiopia's latitude, longitude and four extreme points.
> የኢትዮጵያን ኬክሮስ፣ ኬንትሮስና አራቱን ጽንፈኛ ነጥቦች መግለጽ።

7. Name the six neighbouring countries and the longest and shortest borders.
> ስድስቱን ጎረቤት አገሮችና ረጅሙንና አጭሩን ድንበር መጥቀስ።

8. Give Ethiopia's size and rank, and list advantages and disadvantages of large size.
> የኢትዮጵያን ስፋትና ደረጃ መስጠት፤ የትልቅ ስፋትን ጥቅምና ጉዳት መዘርዘር።

9. Explain why Ethiopia is described as compact and calculate B/C and A/B.
> ኢትዮጵያ ለምን ውሱን (compact) ቅርጽ እንዳላት ማብራራትና B/C እና A/B ማስላት።

10. Name the four geological eras with dates and the main events of each.
> አራቱን የጂኦሎጂ ዘመናት ከቀናቸውና ከዋና ክስተቶቻቸው ጋር መጥቀስ።

11. Explain the Triassic, Jurassic and Cretaceous sequence of Mesozoic rocks.
> የሜሶዞይክ ዐለቶችን የትራይያሲክ፣ ጁራሲክና ክሪቴሸስ ቅደም ተከተል ማብራራት።

12. Describe Tertiary and Quaternary events, including the formation of the Rift Valley.
> የተርሸሪና የኳተርነሪ ክስተቶችን ከስምጥ ሸለቆ አፈጣጠር ጋር መግለጽ።

13. Describe the three physiographic divisions and their sub-divisions, peaks and rivers.
> ሦስቱን የመሬት አቀማመጥ ክፍሎች፣ ንዑስ ክፍሎቻቸውን፣ ጫፎቻቸውንና ወንዞቻቸውን መግለጽ።

14. Describe the Afar Triangle, Main Ethiopian Rift and Chew Bahir Rift.
> የአፋር ሦስት ማዕዘን፣ ዋናው የኢትዮጵያ ስምጥና የጨው ባሕር ስምጥን መግለጽ።

# Key Vocabulary

## ቁልፍ ቃላት

| Term | Meaning | አማርኛ |
|---|---|---|
| **Geography** | Scientific study of the Earth: spatial and temporal variations of physical, biological and human phenomena | ጂኦግራፊ — የምድርን ሳይንሳዊ ጥናት |
| **Scope** | Breadth and depth of a field of study | ወሰን — የጥናት መስክ ስፋትና ጥልቀት |
| **Hydrosphere / Atmosphere** | All water bodies / the layer of gases around Earth | የውሃ ስፈር / የአየር ስፈር |
| **Lithosphere / Biosphere** | Solid outer layer of rocks / all living organisms | የድንጋይ ስፈር / የሕይወት ስፈር |
| **Anthroposphere** | Part of the environment modified by humans | በሰው የተለወጠ አካባቢ |
| **Geomorphology** | Study of landforms and their distribution | የመሬት ቅርጽ ጥናት |
| **Biogeography** | Distribution of plants and animals (zoogeography + phytogeography) | የእጽዋትና እንስሳት ስርጭት ጥናት |
| **Relative location** | Position compared with other features (vicinal and strategic) | አንጻራዊ አቀማመጥ |
| **Absolute location** | Position given by latitude and longitude | ፍጹም (ሥነ ፈለካዊ) አቀማመጥ |
| **Vicinal location** | Position relative to neighbouring countries | ከጎረቤት አገሮች አንጻር ያለ አቀማመጥ |
| **Strategic location** | Position relative to major water bodies and landmasses | ከውሃ አካላትና ከየብስ አንጻር ያለ አቀማመጥ |
| **Landlocked** | Having no coastline on the sea | የባሕር በር የሌለው |
| **Compact shape** | Shape close to a circle; short border for its area | ወደ ክብ የተጠጋ ውሱን ቅርጽ |
| **Hinterland** | Interior land behind a border | ከድንበር ወዲያ ያለ የውስጥ መሬት |
| **Geological time scale** | Division of Earth history into eras and periods | የጂኦሎጂ ጊዜ መለኪያ |
| **Orogenic movement** | Mountain-building movement | ተራራ ፈጣሪ እንቅስቃሴ |
| **Denudation** | Wearing down of land by erosion and weathering | የመሬት መሸርሸር/መላጥ |
| **Peneplanation** | Reduction of land to a nearly flat surface | መሬት ወደ ሚዳነት መዋረድ |
| **Inselberg** | Isolated residual hill left after denudation | ከመሸርሸር የተረፈ ብቸኛ ኮረብታ |

| Term | Meaning | አማርኛ |
|---|---|---|
| **Epeirogenesis** | Slow sinking and rising of large land areas | የመሬት ዝግተኛ መውረድና መውጣት |
| **Transgression / Regression** | Sea advancing onto land / sea retreating | የባሕር መስፋፋት / መሸሽ |
| **Sedimentary rock** | Rock formed from layers of deposited material (sandstone, limestone) | ከደለል የተሰራ ዐለት |
| **Igneous rock** | Rock formed from cooled molten material (basalt) | ከቀለጠ ቁስ የተሰራ ዐለት |
| **Basement complex** | Oldest crystalline rock under all other rocks | ከሁሉም ዐለት በታች ያለ ጥንታዊ ክሪስታል ዐለት |
| **Trappean lava** | Extensive basaltic lava flows of the Tertiary | የተርሸሪ ሰፊ የባሳልት ላቫ |
| **Endogenic / Exogenic forces** | Forces from inside the Earth / at the surface | የውስጥ / የውጭ ኃይሎች |
| **Amba** | Flat-topped mountain / isolated arable highland | አምባ — ጠፍጣፋ አናት ያለ ተራራ |
| **Rift Valley** | Narrow strip of land formed by fractures; a graben | ስምጥ ሸለቆ |
| **Graben / Horst** | Down-dropped block / uplifted block | የወረደ ጥርጊያ / የተነሳ ጥርጊያ |
| **Fumarole** | Vent that releases steam and gases | የእንፋሎት መውጫ ቀዳዳ |
| **a.m.s.l.** | Above mean sea level | ከባሕር ጠለል በላይ |

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# 1.1 Geography: Meaning, Scope and Branches

## 1.1 ጂኦግራፊ፦ ትርጉም፣ ወሰንና ቅርንጫፎች

### 1.1.1 Meaning of Geography

#### 1.1.1 የጂኦግራፊ ትርጉም

ENGLISH

**Origin of the word.** “Geography” comes from ancient Greece and joins two words: **Geo** = earth and **Graphos** = writing. So geography literally means “writing about the earth”.

**Eratosthenes (276–194 B.C.)**, a Greek philosopher, is credited with coining the term. He defined it as “the field of study that deals with the description of the earth.”

**How definitions evolved**

- **Alexander von Humboldt (1769–1859):** a synthesizing discipline that connects the general with the particular through measurement, mapping and a regional emphasis.
- **Concise Oxford Dictionary (1964):** the science of the earth's surfaces.
- **R. Hartshorne (1899–1992):** a branch of knowledge concerned with an accurate, orderly and rational description of distributions on the surface of the earth.
- **M. Yeates (1968):** a science concerned with the rational development and testing of theories that explain and predict the spatial distribution and locations of things and phenomena on the earth's surface.

አማርኛ

> የቃሉ ምንጭ። “ጂኦግራፊ” ከጥንታዊ ግሪክ የመጣ ሲሆን ሁለት ቃላትን ያጣምራል፦ ጂኦ **(Geo)** = ምድር እና ግራፎስ **(Graphos)** = መጻፍ። ስለዚህ ጂኦግራፊ በቀጥታ ሲተረጎም “ስለ ምድር መጻፍ” ማለት ነው።
>
> ኤራቶስቴንስ **(276–194 ዓ.ዓ.)** የተባለው ግሪካዊ ፈላስፋ ቃሉን በመፍጠር ይታወቃል። ጂኦግራፊን “የምድርን መግለጫ የሚመለከት የጥናት መስክ” ሲል ተርጉሞታል።

ትርጓሜዎች እንዴት ተሻሽለዋል

- አሌክሳንደር ፎን ሁምቦልት **(1769–1859)**፦ ጠቅላላውን ከዝርዝሩ ጋር በመለካት፣ በካርታ ስራና በክልል ትኩረት የሚያገናኝ አቀናጂ ዘርፍ።
- ኮንሳይዝ ኦክስፎርድ መዝገበ ቃላት **(1964)**፦ የምድር ገጽ ሳይንስ።
- አር. ሃርትሾርን **(1899–1992)**፦ በምድር ገጽ ላይ ስላሉ ስርጭቶች ትክክለኛ፣ ሥርዓታዊና ምክንያታዊ መግለጫ የሚሰጥ የእውቀት ዘርፍ።
- ኤም. የትስ **(1968)**፦ በምድር ገጽ ላይ ያሉ ነገሮችንና ክስተቶችን የቦታ ስርጭትና አቀማመጥ የሚያስረዱና የሚተነብዩ ንድፈ ሐሳቦችን በምክንያታዊነት የሚያዳብርና የሚፈትን ሳይንስ።

DEFINITION **Contemporary definition / ዘመናዊ ትርጓሜ**

Geography is the scientific study of the Earth that describes and analyses spatial and temporal variations of physical, biological and human phenomena, and their interrelationships and dynamism over the surface of the Earth.

ጂኦግራፊ በምድር ገጽ ላይ ያሉ ተፈጥሯዊ፣ ሕይወታዊና ሰዋዊ ክስተቶችን የቦታና የጊዜ ልዩነቶች፣ እንዲሁም ግንኙነታቸውንና ተለዋዋጭነታቸውን የሚገልጽና የሚተነትን የምድር ሳይንሳዊ ጥናት ነው።

COMMON MISCONCEPTIONS የተለመዱ የተሳሳቱ ግንዛቤዎች

× Humboldt coined the word “geography”.

✓ Eratosthenes (276–194 B.C.) coined it. Humboldt gave a later definition (1769–1859).

> ቃሉን የፈጠረው ኤራቶስቴንስ ነው። ሁምቦልት በኋላ ዘመን ትርጓሜ ሰጥቷል።

× There is one universally accepted definition.

✓ Because geography is dynamic, no single definition is universal; the contemporary definition is the widely accepted one.

> ጂኦግራፊ ተለዋዋጭ በመሆኑ አንድ ሁሉም የሚስማማበት ትርጓሜ የለም፤ ዘመናዊው ትርጓሜ በሰፊው ተቀባይነት ያለው ነው።

## Worked Example 1.1 · Writing a model definition

### ምሳሌ 1.1፡ ናሙና ትርጓሜ መጻፍ

**Question:** Define geography and say where the word comes from. (3 marks)

**Answer:** The word comes from the Greek words geo (earth) and graphos (writing) [1]. Eratosthenes first used it to mean the description of the earth [1]. Today geography is the scientific study of the Earth that describes and analyses spatial and temporal variations of physical, biological and human phenomena and their interrelationships [1].

> መልስ፦ ቃሉ ከግሪክ ቃላት ጂኦ (ምድር) እና ግራፎስ (መጻፍ) የተገኘ ነው። ኤራቶስቴንስ የምድር መግለጫ ለማለት ተጠቀመበት። ዛሬ ጂኦግራፊ የምድርን ተፈጥሯዊ፣ ሕይወታዊና ሰዋዊ ክስተቶች የቦታና ጊዜ ልዩነቶችና ግንኙነቶች የሚተነትን ሳይንሳዊ ጥናት ነው።

KEY TAKEAWAYS **Meaning of geography / የጂኦግራፊ ትርጉም**

- Geo = earth, graphos = writing; coined by Eratosthenes (276–194 B.C.).
  ጂኦ = ምድር፣ ግራፎስ = መጻፍ፤ በኤራቶስቴንስ የተፈጠረ።
- Four scholars to remember: Humboldt (1769), Oxford (1964), Hartshorne (1899–1992), Yeates (1968).
  አራት ምሁራን፦ ሁምቦልት፣ ኦክስፎርድ፣ ሃርትሾርን፣ የትስ።
- Key words of the modern definition: spatial, temporal, physical, biological, human, interrelationships.
  ቁልፍ ቃላት፦ የቦታ፣ የጊዜ፣ ተፈጥሯዊ፣ ሕይወታዊ፣ ሰዋዊ፣ ግንኙነት።

## 1.1.2 The Scope of Geography

## 1.1.2 የጂኦግራፊ ወሰን

### ENGLISH

**Scope** means the breadth and depth of a field of study. The scope of geography is very wide and dynamic: it changes with new discoveries and ideas.

Geography studies physical and human features, their causes, effects and interactions. These are grouped in five spheres:

- **Hydrosphere** – all bodies of water (oceans, rivers, lakes).
- **Biosphere** – all living organisms on Earth's surface.
- **Atmosphere** – the layer of gases surrounding the planet.
- **Lithosphere** – the solid outer layer of rocks.
- **Anthroposphere** – the part of the environment modified by humans.

### Major focus areas

- The Earth's position in the universe and its movements.
- Earth's physical features, the forces shaping them and their spatial and temporal variations.
- Interrelationships and interdependence between people and their natural environment.
- Atmospheric conditions, weather and the distribution of climate.
- Earth's materials and diverse landforms.
- Major human economic activities and their environmental impacts.

Geography also addresses economic, social, political and ecological problems and uses modern technologies such as **GPS** and **GIS**. It is a **holistic discipline** because it joins physical, biotic and human environments.

### አማርኛ

> ወሰን **(scope)** ማለት የአንድ የጥናት መስክ ስፋትና ጥልቀት ማለት ነው። የጂኦግራፊ ወሰን እጅግ ሰፊና ተለዋዋጭ ነው፤ በአዳዲስ ግኝቶችና ሐሳቦች ይለወጣል።
>
> ጂኦግራፊ ተፈጥሯዊና ሰዋዊ ገጽታዎችን፣ ምክንያቶቻቸውን፣ ውጤቶቻቸውንና ግንኙነቶቻቸውን ያጠናል። እነዚህ በአምስት ስፈሮች ይመደባሉ፦

- የውሃ ስፈር **(Hydrosphere)** – ሁሉም የውሃ አካላት (ውቅያኖሶች፣ ወንዞች፣ ሐይቆች)።
- የሕይወት ስፈር **(Biosphere)** – በምድር ገጽ ላይ ያሉ ሕያዋን ፍጥረታት በሙሉ።
- የአየር ስፈር **(Atmosphere)** – ፕላኔቷን የከበበ የጋዝ ሽፋን።
- የድንጋይ ስፈር **(Lithosphere)** – ደረቅ የውጭ የዐለት ሽፋን።
- የሰው ስፈር **(Anthroposphere)** – በሰው የተለወጠ የአካባቢ ክፍል።

### ዋና ትኩረት ነጥቦች

- የምድር በጠፈር ውስጥ ያላት ቦታና እንቅስቃሴዎቿ።
- የምድር ተፈጥሯዊ ገጽታዎች፣ የሚቀርጿቸው ኃይሎችና የቦታና ጊዜ ልዩነቶቻቸው።
- በሰዎችና በተፈጥሮ አካባቢያቸው መካከል ያለ ግንኙነትና ጥገኝነት።
- የከባቢ አየር ሁኔታ፣ የአየር ጠባይና የአየር ንብረት ስርጭት።
- የምድር ቁሶችና የተለያዩ የመሬት ቅርጾች።

- ዋና ዋና የሰው ኢኮኖሚያዊ ተግባራትና በአካባቢ ላይ ያላቸው ተጽዕኖ።

ጂኦግራፊ ኢኮኖሚያዊ፣ ማኅበራዊ፣ ፖለቲካዊና ሥነ ምህዳራዊ ችግሮችንም ይመለከታል፤ እንደ **GPS** እና **GIS** ያሉ ዘመናዊ ቴክኖሎጂዎችን ይጠቀማል። ተፈጥሯዊ፣ ሕይወታዊና ሰዋዊ አካባቢዎችን ስለሚያዋህድ ሁለንተናዊ **(holistic)** ዘርፍ ነው።

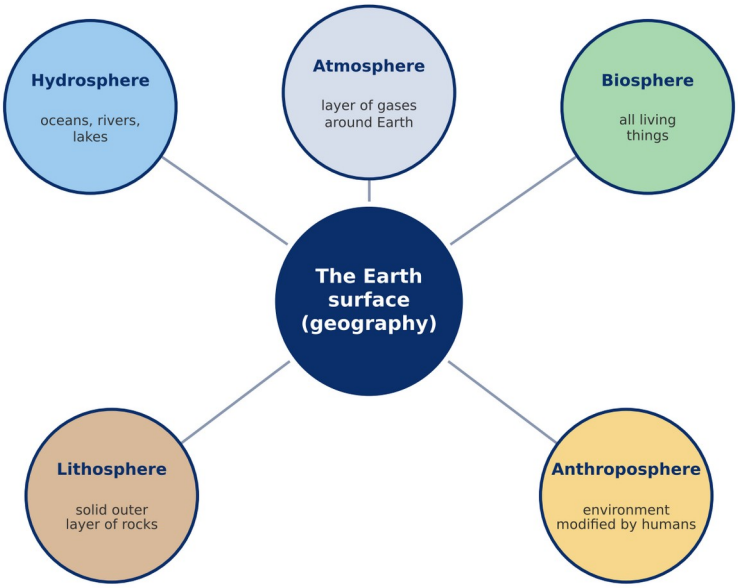



**Figure 1.1** *The five spheres studied in geography.*
ስዕል 1.1፦ በጂኦግራፊ የሚጠኑ አምስቱ ስፌሮች።

**COMMON MISCONCEPTIONS** የተለመዱ የተሳሳቱ ግንዛቤዎች

× Geography studies only the physical Earth (mountains, rivers).
✓ Geography is holistic: it studies physical, biological and human phenomena and how they interact.

ጂኦግራፊ ተራሮችንና ወንዞችን ብቻ አያጠናም፤ ተፈጥሯዊ፣ ሕይወታዊና ሰዋዊ ክስተቶችንና ግንኙነታቸውን ያጠናል።

× The anthroposphere is part of nature untouched by people.
✓ The anthroposphere is the environment MODIFIED by humans (farms, cities, roads).

የሰው ስፌር በሰው የተለወጠ አካባቢ ነው (እርሻ፣ ከተማ፣ መንገድ)።

### Worked Example 1.2 · Matching features to spheres

### ምሳሌ 1.2፡ ገጽታዎችን ከስፌሮች ጋር ማዛመድ

**Question:** Which sphere do these belong to? (a) Lake Tana (b) a coffee forest in Jimma (c) the Ethiopian highland rocks (d) a terraced hillside farm (e) the air above Addis Ababa.

**Answer:** (a) hydrosphere; (b) biosphere; (c) lithosphere; (d) anthroposphere (land modified by people); (e) atmosphere.

መልስ፦ (ሀ) የውሃ ስፌር፤ (ለ) የሕይወት ስፌር፤ (ሐ) የድንጋይ ስፌር፤ (መ) የሰው ስፌር (በሰው የተለወጠ መሬት)፤ (ሠ) የአየር ስፌር።

**KEY TAKEAWAYS** **Scope of geography / የጂኦግራፊ ወሰን**

- Five spheres: hydro-, atmo-, litho-, bio-, anthropo-sphere.
  አምስት ስፈሮች።
- Scope is wide and dynamic; geography is a holistic discipline that uses GPS and GIS.
  ወሰኑ ሰፊና ተለዋዋጭ ነው፤ ሁለንተናዊ ዘርፍ ነው።

## 1.1.3 Branches of Geography

### 1.1.3 የጂኦግራፊ ቅርንጫፎች

**ENGLISH**

Geography has **two main branches**: **Physical Geography** (natural features) and **Human Geography** (people and their activities).

**አማርኛ**

> ጂኦግራፊ ሁለት ዋና ቅርንጫፎች አሉት፦ ተፈጥሯዊ ጂኦግራፊ (የተፈጥሮ ገጽታዎች) እና ሰዋዊ ጂኦግራፊ (ሰዎችና ተግባራቸው)።

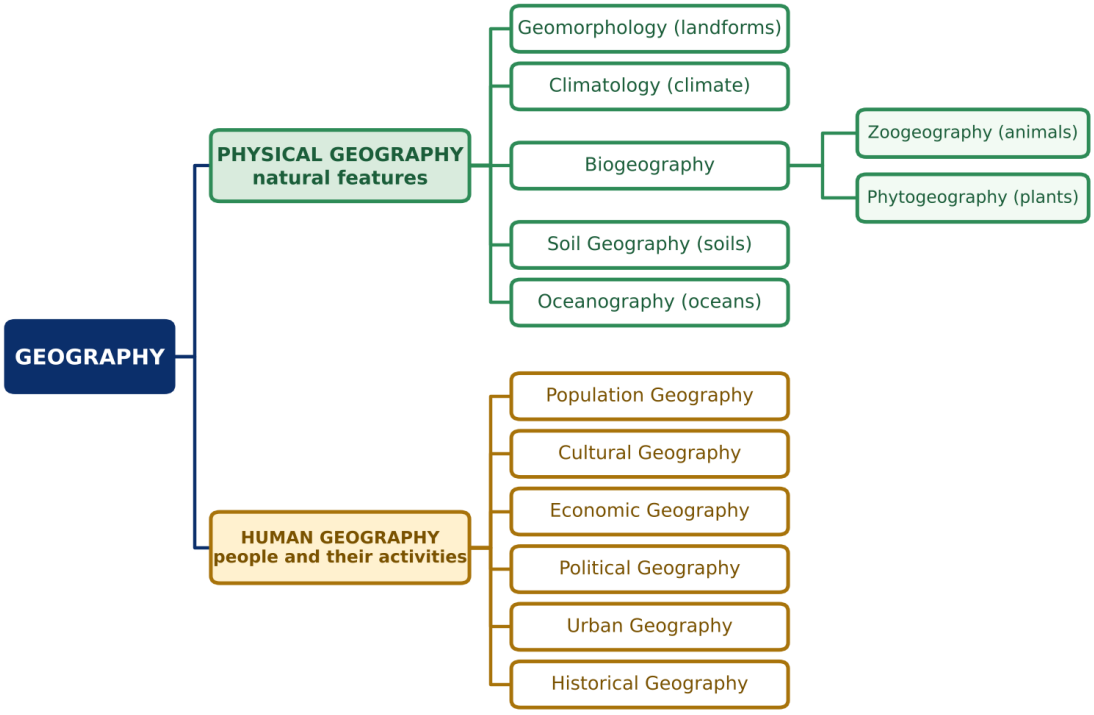



**Figure 1.2** *The branches of geography.*
ስዕል 1.2፦ የጂኦግራፊ ቅርንጫፎች።

| Branch | What it studies | አማርኛ |
|---|---|---|
| **Geomorphology** | Landforms and their distribution | የመሬት ቅርጾችና ስርጭታቸው |
| **Climatology** | Patterns of climate | የአየር ንብረት ሥርዓት |
| **Biogeography** | Distribution of plants and animals | የእጽዋትና እንስሳት ስርጭት |

| Branch | What it studies | አማርኛ |
|---|---|---|
| Zoogeography | Distribution of animals | የእንስሳት ስርጭት |
| Phytogeography | Distribution of plants | የእጽዋት ስርጭት |
| Soil Geography | Distribution of soils | የአፈር ስርጭት |
| Oceanography | Oceans | ውቅያኖሶች |
| Population Geography | Distribution and characteristics of human populations | የሕዝብ ስርጭትና ባሕርያት |
| Cultural Geography | Spatial distribution and diffusion of cultural traits | የባህል ባሕርያት ስርጭትና መስፋፋት |
| Economic Geography | Location, distribution and spatial organization of economic activities | የኢኮኖሚ ተግባራት አቀማመጥና አደረጃጀት |
| Political Geography | Spatial aspects of politics and political organization | የፖለቲካ የቦታ ገጽታዎች |
| Urban Geography | Spatial structure and organization of cities | የከተሞች መዋቅርና አደረጃጀት |
| Historical Geography | Geographical phenomena in past times | ያለፉ ዘመናት ጂኦግራፊያዊ ክስተቶች |

**COMMON MISCONCEPTIONS** የተለመዱ የተሳሳቱ ግንዛቤዎች

× Oceanography belongs to human geography.
✓ Oceanography is a sub-branch of PHYSICAL geography (with geomorphology, climatology, biogeography, soil geography).
ውቅያኖስ ጥናት የተፈጥሯዊ ጂኦግራፊ ቅርንጫፍ ነው።

× Zoogeography studies plants.
✓ Zoogeography = animals; phytogeography = plants. Together they form biogeography.
ዞኦጂኦግራፊ እንስሳትን፤ ፋይቶጂኦግራፊ እጽዋትን ያጠናል።

× Historical geography is a branch of physical geography.
✓ It is a sub-branch of HUMAN geography.
ታሪካዊ ጂኦግራፊ የሰዋዊ ጂኦግራፊ ቅርንጫፍ ነው።

### Worked Example 1.3 · Which branch?

### ምሳሌ 1.3፡ የትኛው ቅርንጫፍ?

**Question:** Name the branch that studies (a) why wheat is grown mostly on the Arsi plateau, (b) the distribution of the Walia ibex, (c) how Addis Ababa is laid out, (d) the shape and distribution of mountains, (e) how Ethiopian borders were drawn in the past.

**Answer:** (a) economic geography; (b) zoogeography (biogeography); (c) urban geography; (d) geomorphology; (e) political or historical geography.

> መልስ፦ (ሀ) ኢኮኖሚ ጂኦግራፊ፤ (ለ) ዞኦጂኦግራፊ (ባዮጂኦግራፊ)፤ (ሐ) የከተማ ጂኦግራፊ፤ (መ) ጂኦሞርፎሎጂ፤ (ሠ) የፖለቲካ ወይም ታሪካዊ ጂኦግራፊ።

**KEY TAKEAWAYS** **Branches / ቅርንጫፎች**

- Physical (5): geomorphology, climatology, biogeography (zoo + phyto), soil, oceanography.
  ተፈጥሯዊ፦ ጂኦሞርፎሎጂ፣ ክሊማቶሎጂ፣ ባዮጂኦግራፊ፣ አፈር፣ ውቅያኖስ።
- Human (6): population, cultural, economic, political, urban, historical.
  ሰዋዊ፦ ሕዝብ፣ ባህል፣ ኢኮኖሚ፣ ፖለቲካ፣ ከተማ፣ ታሪካዊ።

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# 1.2 Location, Size and Shape of Ethiopia

## 1.2 የኢትዮጵያ አቀማመጥ፣ ስፋትና ቅርጽ

### 1.2.1 Location of Ethiopia

### 1.2.1 የኢትዮጵያ አቀማመጥ

ENGLISH

Location describes the position of a place in relation to its surroundings. There are two main kinds:

- **Relative location** – position compared with other features. It has two parts: **vicinal** (neighbouring countries) and **strategic / natural / global** (water bodies and landmasses).
- **Absolute (astronomical) location** – position expressed in latitude and longitude.

አማርኛ

> አቀማመጥ የአንድን ቦታ ከአካባቢው አንጻር ያለውን ቦታ ይገልጻል። ሁለት ዋና ዓይነቶች አሉት፦
>
> - አንጻራዊ አቀማመጥ – ከሌሎች ገጽታዎች ጋር ሲነጻጸር ያለ ቦታ። ሁለት ክፍሎች አሉት፦ አጎራባች **(vicinal)** (ጎረቤት አገሮች) እና ስትራቴጂያዊ / ተፈጥሯዊ / ዓለም አቀፋዊ (የውሃ አካላትና የየብስ ክፍሎች)።
> - ፍጹም (ሥነ ፈለካዊ) አቀማመጥ – በኬክሮስና በኬንትሮስ የሚገለጽ ቦታ።

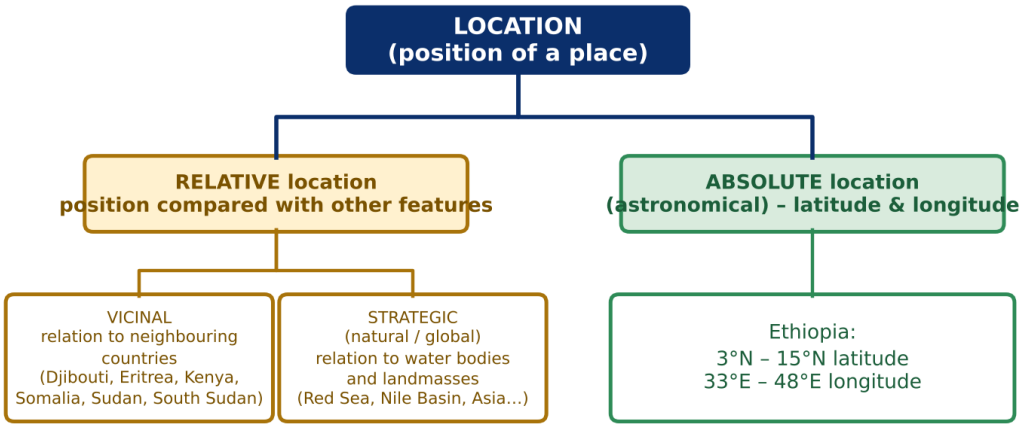



**Figure 1.3** *Types of location.*
ስዕል 1.3፦ የአቀማመጥ ዓይነቶች።

ENGLISH

#### Vicinal location

አማርኛ

#### አጎራባች አቀማመጥ

ENGLISH

Ethiopia is a **landlocked** country bordered by **six** nations: Djibouti, Eritrea, Kenya, Somalia, Sudan and South Sudan.

- Total boundary length: **5,260 km**.

- Longest boundary: **Somalia (1,600 km)**.
- Shortest boundary: **Djibouti (310 km)**.

አማርኛ

> ኢትዮጵያ የባሕር በር የሌላት አገር ስትሆን ስድስት አገሮችን ትዋሰናለች፦ ጅቡቲ፣ ኤርትራ፣ ኬንያ፣ ሶማሊያ፣ ሱዳንና ደቡብ ሱዳን።

- ጠቅላላ የድንበር ርዝመት፦ **5,260** ኪ.ሜ።
- ረጅሙ ድንበር፦ ከሶማሊያ ጋር **(1,600** ኪ.ሜ)።
- አጭሩ ድንበር፦ ከጅቡቲ ጋር **(310** ኪ.ሜ)።

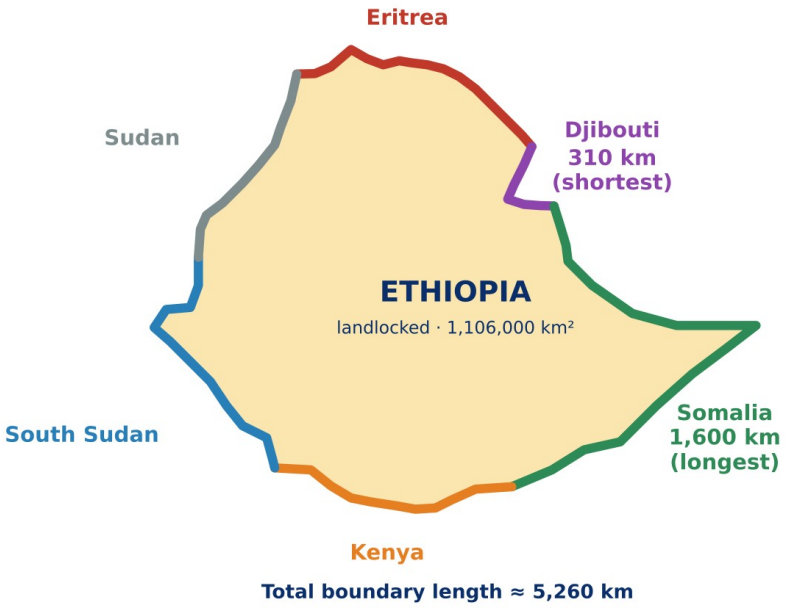



**Figure 1.4** *Ethiopia's six neighbours (outline schematic; lengths from the textbook).*
ስዕል 1.4፦ የኢትዮጵያ ስድስት ጎረቤቶች (ንድፍ፤ ርዝመቶች ከመማሪያ መጽሐፍ)።

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ENGLISH

## Strategic location

አማርኛ

## ስትራቴጂያዊ አቀማመጥ

ENGLISH

- In north-eastern (Horn of) Africa.
- South-west of the Asian continent.
- Within the Nile Basin.
- South of Europe and south of the Mediterranean Sea.
- North-west of the Indian Ocean.
- South-west of the Red Sea.

አማርኛ

- በሰሜን ምስራቅ አፍሪካ (የአፍሪካ ቀንድ)።
- ከእስያ አህጉር ደቡብ ምዕራብ።

- በአባይ ተፋሰስ ውስጥ።
- ከአውሮፓና ከሜዲትራኒያን ባሕር በደቡብ።
- ከህንድ ውቅያኖስ ሰሜን ምዕራብ።
- ከቀይ ባሕር ደቡብ ምዕራብ።

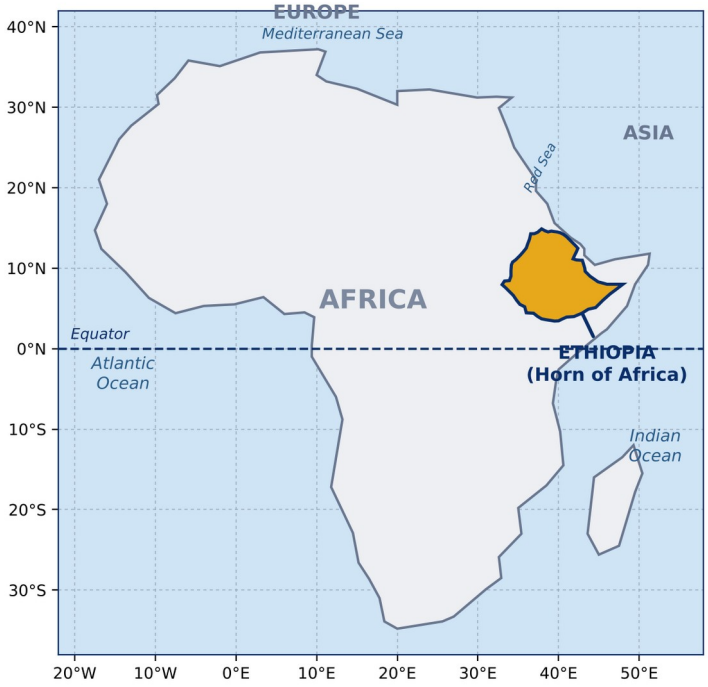



**Figure 1.5** *Ethiopia in Africa.*
ስዕል 1.5፦ ኢትዮጵያ በአፍሪካ።

ENGLISH

## Absolute location

አማርኛ

## ፍጹም አቀማመጥ

ENGLISH

Ethiopia lies between **3°N and 15°N** latitude and **33°E and 48°E** longitude.

- **North:** northern tip of Tigray (about 15°N).
- **South:** Moyale (Borena) (about 3°N).
- **West:** Akobo (Gambella) (about 33°E).
- **East:** tip of the Ogaden (about 48°E).

Latitude span = 15° − 3° = **12°**; longitude span = 48° − 33° = **15°**.

አማርኛ

> ኢትዮጵያ በ **3°N** እና **15°N** ኬክሮስ እንዲሁም በ **33°E** እና **48°E** ኬንትሮስ መካከል ትገኛለች።

- ሰሜን፦ የትግራይ ሰሜናዊ ጫፍ (ወደ 15°N)።
- ደቡብ፦ ሞያሌ (ቦረና) (ወደ 3°N)።
- ምዕራብ፦ አኮቦ (ጋምቤላ) (ወደ 33°E)።
- ምስራቅ፦ የኦጋዴን ጫፍ (ወደ 48°E)።

የኬክሮስ ስፋት = 15° − 3° = **12°**፤ የኬንትሮስ ስፋት = 48° − 33° = **15°**።

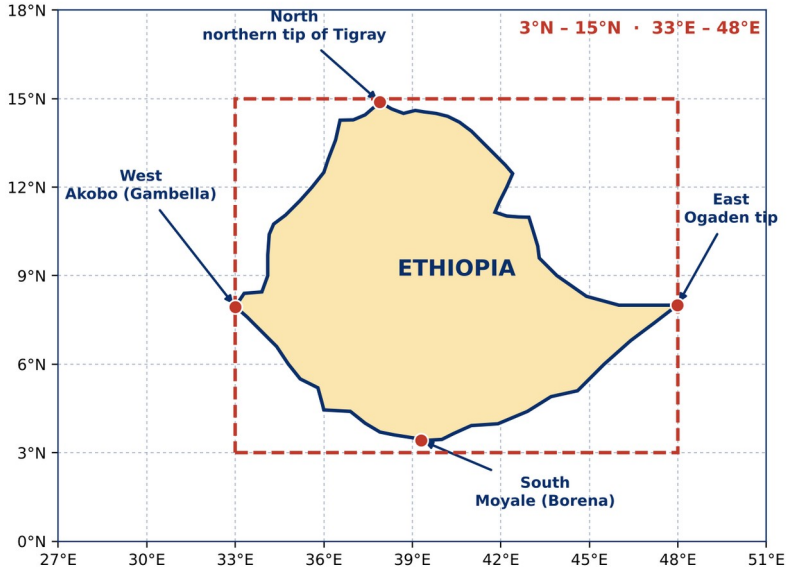



**Figure 1.6** *Absolute location and extreme points of Ethiopia.*
ስዕል 1.6፦ የኢትዮጵያ ፍጹም አቀማመጥና ጽንፈኛ ነጥቦች።

**COMMON MISCONCEPTIONS** የተለመዱ የተሳሳቱ ግንዛቤዎች

× Ethiopia has a coastline on the Red Sea.
✓ Ethiopia is LANDLOCKED; it has six neighbours but no sea coast.
ኢትዮጵያ የባሕር በር የላትም፤ ስድስት ጎረቤቶች አሏት።

× Ethiopia's longest border is with Sudan.
✓ The longest border is with Somalia (1,600 km); the shortest is with Djibouti (310 km).
ረጅሙ ድንበር ከሶማሊያ (1,600 ኪ.ሜ)፤ አጭሩ ከጅቡቲ (310 ኪ.ሜ)።

× "North of the Equator" is relative location.
✓ Latitude and longitude values are ABSOLUTE location. Relative location compares with other places.
ኬክሮስና ኬንትሮስ ፍጹም አቀማመጥ ናቸው፤ አንጻራዊ ከሌሎች ቦታዎች ጋር ማወዳደር ነው።

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## Worked Example 1.4 · Classify the location statement

### ምሳሌ 1.4፡ የአቀማመጥ አረፍተ ነገሮችን መደብ

**Question:** Classify each as vicinal, strategic or absolute: (a) "Ethiopia shares a 310 km border with Djibouti." (b) "Ethiopia lies between 33°E and 48°E." (c) "Ethiopia is within the Nile Basin." (d) "Ethiopia is bordered by Kenya in the south."

**Answer:** (a) vicinal; (b) absolute; (c) strategic; (d) vicinal.

መልስ፦ (ሀ) አጎራባች፤ (ለ) ፍጹም፤ (ሐ) ስትራቴጂያዊ፤ (መ) አጎራባች።

### Worked Example 1.5 · Spans and extremes

**ምሳሌ 1.5፡ ስፋቶችና ጽንፎች**

**Question:** The northernmost point is at 15°N and the southernmost at 3°N. If 1° of latitude ≈ 111 km, estimate the north–south extent.

$$12^\circ \times 111 \text{ km} = 1{,}332 \text{ km} \quad (\text{about } 1{,}300 \text{ km})$$

**Interpretation:** the east–west span (15°) is similar to the north–south span (12°), which supports the idea of a compact shape.

> መልስ፦ የሰሜን–ደቡብ ርቀት ≈ 12° × 111 = 1,332 ኪ.ሜ። የምስራቅ–ምዕራብ ስፋት (15°) ከሰሜን–ደቡብ (12°) ጋር ተቀራራቢ ነው፤ ይህም ውሱን ቅርጽን ይደግፋል።

**KEY TAKEAWAYS** **Location / አቀማመጥ**

- Landlocked; six neighbours: Djibouti, Eritrea, Kenya, Somalia, Sudan, South Sudan.
  የባሕር በር የሌላት፤ ስድስት ጎረቤቶች።
- Boundary 5,260 km; longest Somalia 1,600 km; shortest Djibouti 310 km.
  ድንበር 5,260 ኪ.ሜ፤ ረጅም ሶማሊያ፤ አጭር ጅቡቲ።
- Absolute: 3°N–15°N, 33°E–48°E. Extremes: Tigray (N), Moyale (S), Akobo (W), Ogaden (E).
  ፍጹም፦ 3°N–15°N፣ 33°E–48°E።

## 1.2.2 Size of Ethiopia

## 1.2.2 የኢትዮጵያ ስፋት

**ENGLISH**

- Total area: **1,106,000 km²**.
- **Tenth largest** country in Africa.
- **0.7 %** of the world's land area and **3.6 %** of Africa's land mass.
- Larger than most of its neighbours (except Sudan) and the **largest country in the Horn of Africa**.

### Advantages of large size

- Diverse agro-ecological zones, so many kinds of animals and plants.
- Large amount of arable land.
- Great variety of mineral resources.
- Home to diverse ethnic groups.

### Disadvantages of large size

- Expensive administration.
- Needs a large army for defence.
- Needs strong finance for roads and other infrastructure.
- Hard to administer and to integrate socially and economically.

አማርኛ

- ጠቅላላ ስፋት፦ **1,106,000** ካሬ ኪ.ሜ።
- በአፍሪካ አሥረኛዋ ትልቅ አገር።
- ከዓለም የየብስ ስፋት **0.7 %** እና ከአፍሪካ የየብስ ስፋት **3.6 %**።
- ከጎረቤቶቿ አብዛኞቹ (ከሱዳን በስተቀር) ትበልጣለች፤ በአፍሪካ ቀንድ ትልቋ አገር ናት።

የትልቅ ስፋት ጥቅሞች

- የተለያዩ የግብርና ሥነ ምህዳር ዞኖች፤ ብዙ የእንስሳትና የእጽዋት ዓይነቶች።
- ብዙ የሚታረስ መሬት።
- የተለያዩ የማዕድን ሀብቶች።
- የተለያዩ ብሔረሰቦች መኖሪያ።

የትልቅ ስፋት ጉዳቶች

- ውድ የአስተዳደር ወጪ።
- ለመከላከያ ትልቅ ሠራዊት ያስፈልጋል።
- ለመንገድና ለሌሎች መሠረተ ልማቶች ጠንካራ ገንዘብ ያስፈልጋል።
- ለማስተዳደርና ማኅበራዊና ኢኮኖሚያዊ ውህደት ለማምጣት አስቸጋሪ።

**COMMON MISCONCEPTIONS** የተለመዱ የተሳሳቱ ግንዛቤዎች

× Ethiopia is the largest country in Africa.
✓ Ethiopia is the TENTH largest in Africa, though the largest in the Horn of Africa.
ኢትዮጵያ በአፍሪካ አሥረኛዋ ናት፤ በአፍሪካ ቀንድ ግን ትልቋ ናት።

× Large size is only an advantage.
✓ It brings both advantages (resources, diversity) and disadvantages (cost of administration, defence, infrastructure).
ትልቅ ስፋት ጥቅምም ጉዳትም አለው።

### Worked Example 1.6 · Using the size percentages

ምሳሌ 1.6፡ የስፋት መቶኛዎችን መጠቀም

**Question:** Ethiopia's area (1,106,000 km²) is 3.6 % of Africa. Estimate Africa's area.

$$\text{Africa} \approx 1{,}106{,}000 \div 0.036 \approx 30.7 \text{ million km}^2$$

**Check:** Africa is roughly 30 million km², so the figures are consistent.

> መልስ፦ የአፍሪካ ስፋት ≈ 1,106,000 ÷ 0.036 ≈ 30.7 ሚሊዮን ካ.ኪ.ሜ። ይህ ከታወቀው የአፍሪካ ስፋት (ወደ 30 ሚሊዮን) ጋር ይስማማል።

**KEY TAKEAWAYS** **Size / ስፋት**

- 1,106,000 km$^2$; 10th in Africa; 0.7 % of world land; 3.6 % of Africa; largest in the Horn.
  1,106,000 ካ.ኪ.ሜ፤ በአፍሪካ 10ኛ።
- Advantages: ecological variety, arable land, minerals, diverse peoples. Disadvantages: costly administration, army, infrastructure.
  ጥቅም፦ ልዩነት፣ መሬት፣ ማዕድን። ጉዳት፦ ወጪ።

## 1.2.3 Shape of Ethiopia

## 1.2.3 የኢትዮጵያ ቅርጽ

**ENGLISH**

Countries may be **compact, elongated, prorupt** (with a narrow arm) or fragmented. Ethiopia is **more or less compact (circular)** because its extreme north–south and east–west spans are comparable.

### Three indicators of compactness

- **Boundary–Circumference ratio (B/C):** B = actual boundary length; C = circumference of a circle with the same area as the country. The closer to **1**, the more compact.
- **Area–Boundary ratio (A/B):** area per kilometre of boundary. The **larger** the value, the more compact.
- **Actual area ÷ area of the smallest circle that contains the country (A/A′):** the closer to **1**, the more compact.

**Main advantage of a compact shape:** military. A short border for the area enclosed makes defence easier and more efficient.

**አማርኛ**

> አገሮች ውሱን **(compact)**፣ ረዘም ያሉ **(elongated)**፣ ክንድ ያላቸው **(prorupt)** ወይም የተበታተኑ ሊሆኑ ይችላሉ። የኢትዮጵያ የሰሜን–ደቡብና የምስራቅ–ምዕራብ ርቀቶች ተቀራራቢ ስለሆኑ ይበልጥ ወይም ባነሰ ውሱን (ክብ መሰል) ናት።

**ሦስት የውሱንነት መለኪያዎች**

- የድንበር ከክብ ዙሪያ ሬሾ **(B/C)**፦ B = የድንበር ርዝመት፤ C = ከአገሪቱ ጋር እኩል ስፋት ያለው ክብ ዙሪያ። ወደ **1** በቀረበ መጠን ይበልጥ ውሱን ነው።
- የስፋት ከድንበር ሬሾ **(A/B)**፦ በአንድ ኪሎ ሜትር ድንበር የሚወድቅ ስፋት። ዋጋው በትልቁ መጠን ይበልጥ ውሱን ነው።
- ትክክለኛ ስፋት ÷ አገሪቱን የሚይዘው ትንሹ ክብ ስፋት **(A/A′)**፦ ወደ **1** በቀረበ መጠን ይበልጥ ውሱን ነው።

> የውሱን ቅርጽ ዋና ጥቅም፦ ወታደራዊ ነው። ለተከበበው ስፋት አጭር ድንበር መከላከልን ቀላልና ቀልጣፋ ያደርጋል።

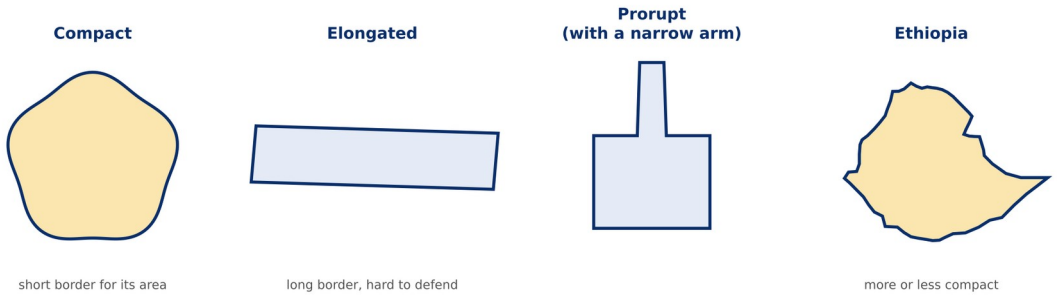



**Figure 1.7** *Country shapes compared.*
ስዕል 1.7፦ የአገር ቅርጾች ንጽጽር።

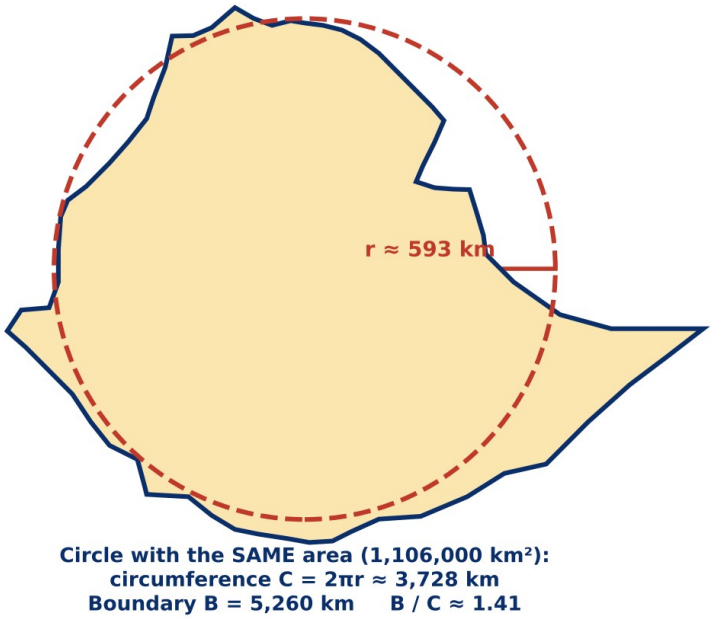



**Figure 1.8** *Ethiopia compared with a circle of equal area (r ≈ 593 km). Outline schematic; values from the notes.*
ስዕል 1.8፦ ኢትዮጵያ ከእኩል ስፋት ክብ ጋር ስትነጻጸር።

### Worked Example 1.7 · Calculate B/C for Ethiopia

### ምሳሌ 1.7፡ ለኢትዮጵያ B/C ማስላት

**Given:** A = 1,106,000 km²; B = 5,260 km. Find B/C.

$$\text{Step 1: radius of the equal-area circle } r = \sqrt{(A \div \pi)} = \sqrt{(1{,}106{,}000 \div 3.1416)} = \sqrt{352{,}053} \approx 593.3 \text{ km}$$

$$\text{Step 2: circumference } C = 2\pi r = 2 \times 3.1416 \times 593.3 \approx 3{,}728 \text{ km}$$

$$\text{Step 3: } B/C = 5{,}260 \div 3{,}728 \approx 1.411$$

**Conclusion:** B/C ≈ 1.411, i.e. Ethiopia's border is about 41 % longer than the perfect-circle minimum. Ethiopia is fairly compact (a perfect circle would give 1.0).

> መልስ፦ r = √(1,106,000 ÷ π) ≈ 593.3 ኪ.ሜ፤ C = 2πr ≈ 3,728 ኪ.ሜ፤ B/C = 5,260 ÷ 3,728 ≈ **1.411**። ይህም የኢትዮጵያ ድንበር ከፍጹም ክብ ወደ 41 % እንደሚረዝም ያሳያል፤ ስለዚህ ኢትዮጵያ በአንጻራዊነት ውሱን ናት።

### Worked Example 1.8 · Calculate A/B and interpret

### ምሳሌ 1.8፡ A/B ማስላትና መተርጎም

**Given:** A = 1,106,000 km²; B = 5,260 km.

$$A/B = 1{,}106{,}000 \div 5{,}260 \approx 210.3 \text{ km}^2 \text{ per km}$$

**Interpretation:** each 1 km of border defended protects about 210 km² of hinterland. A bigger number means a more compact, more defensible country.

> መልስ፦ A/B = 1,106,000 ÷ 5,260 ≈ **210** ካ.ኪ.ሜ በአንድ ኪ.ሜ። አንድ ኪሎ ሜትር ድንበር ሲጠበቅ ወደ 210 ካ.ኪ.ሜ የውስጥ መሬት ይጠበቃል።

### Worked Example 1.9 · Compare two countries

### ምሳሌ 1.9፡ ሁለት አገሮችን ማወዳደር

**Question:** Country X: A = 400,000 km², B = 2,000 km. Country Y: A = 400,000 km², B = 3,200 km. Which is more compact?

$$A/B \text{ for X} = 400{,}000 \div 2{,}000 = 200 \qquad A/B \text{ for Y} = 400{,}000 \div 3{,}200 = 125$$

**Answer:** X is more compact (higher A/B; a shorter border for the same area).

> መልስ፦ X ይበልጥ ውሱን ነው (A/B ከፍ ያለ፤ ለእኩል ስፋት አጭር ድንበር)።

**COMMON MISCONCEPTIONS** የተለመዱ የተሳሳቱ ግንዛቤዎች

× A B/C ratio of 1.411 means Ethiopia is not compact.

✓ B/C is compared with 1.0 (a perfect circle). 1.411 shows moderate compactness: about 41 % above the circle.

B/C ከ1.0 ጋር ይነጻጸራል፤ 1.411 መካከለኛ ውሱንነት ያሳያል (ከክብ 41 % ከፍ ያለ)።

× A higher A/B means a weaker border.

✓ A higher A/B means MORE area per km of border, i.e. MORE compact and easier to defend.

ከፍ ያለ A/B ማለት ይበልጥ ውሱን፣ ለመከላከል ቀላል ማለት ነው።

× Compactness mainly helps the economy.

✓ Its main advantage in the notes is MILITARY (defence).

የውሱን ቅርጽ ዋና ጥቅም ወታደራዊ ነው።

**KEY TAKEAWAYS** **Shape / ቅርጽ**

- Ethiopia: more or less compact because N–S and E–W spans are similar.
  ኢትዮጵያ ውሱን ናት።
- B/C → closer to 1 = compact (Ethiopia 1.411). A/B → larger = compact (Ethiopia ≈ 210 km² per km).
  B/C ወደ 1፤ A/B ትልቅ = ውሱን።
- Advantage of compactness: military/defence.
  ጥቅም፦ ወታደራዊ።

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# 1.3 Geological History of Ethiopia

## 1.3 የኢትዮጵያ የጂኦሎጂ ታሪክ

### 1.3.1 The Geological Processes

### 1.3.1 የጂኦሎጂ ሂደቶች

ENGLISH

Scientists divide Earth's history on the **geological time scale** into four major eras. Eras differ in the relative positions of the continents, the climate and the dominant life forms.

አማርኛ

> ሳይንቲስቶች የምድርን ታሪክ በጂኦሎጂ ጊዜ መለኪያ በአራት ታላላቅ ዘመናት ይከፍሉታል። ዘመናት በአህጉራት አንጻራዊ አቀማመጥ፣ በአየር ንብረትና በዋና ሕይወት ዓይነቶች ይለያያሉ።

| Era | Time (years ago) | Main process / rock | Life forms |
|---|---|---|---|
| **Precambrian** ፕሪካምብሪያን | 4.5 billion – 600 million | Orogenic movements, volcanism, folded mountains; basement complex | First life (amoeba, jellyfish) |
| **Paleozoic** ፓሊዮዞይክ | 600 – 250 million | Denudation → peneplanation; inselbergs; gap in rock formation | Invertebrates |
| **Mesozoic** ሜሶዞይክ | 250 – 70 million | Subsidence & uplift; sedimentary rocks (Adigrat, Hintalo, Upper sandstone) | Reptiles (dinosaurs); at the end mammals, birds, flowering plants |
| **Cenozoic** ሴኖዞይክ | 70 million – present | Uplift, cracking, trap basalt, highlands, Rift Valley, volcanoes | Modern life; human evolution (Quaternary) |

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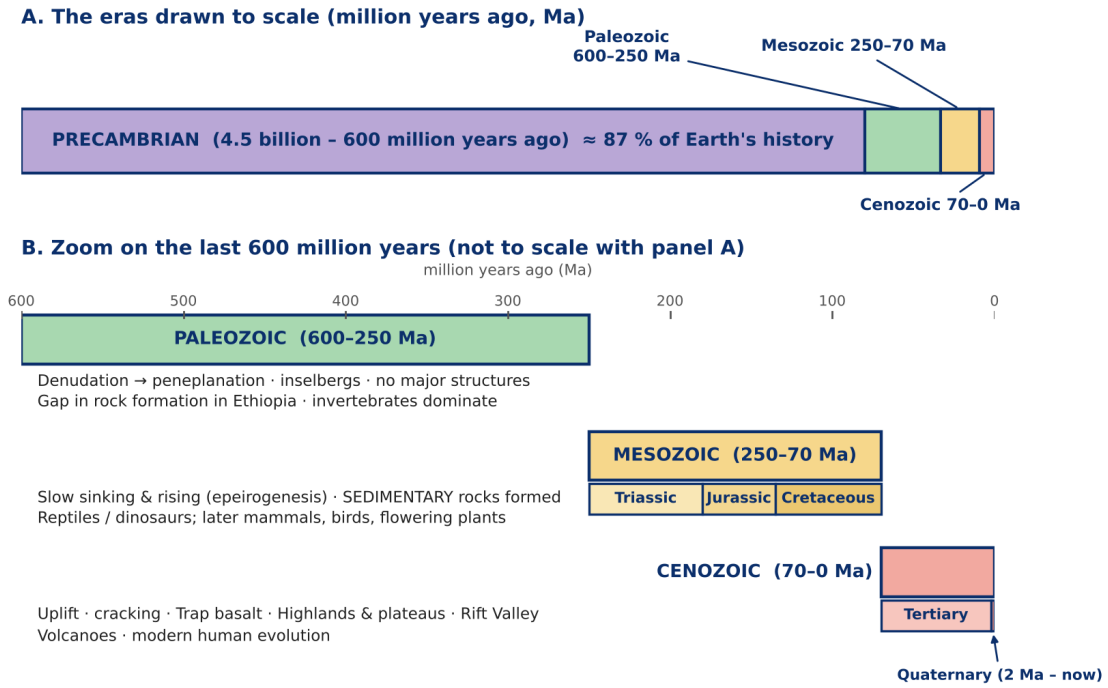



**Figure 1.9** *The geological time scale (eras drawn to scale, then zoomed).*
ስዕል 1.9፦ የጂኦሎጂ ጊዜ መለኪያ።

ENGLISH

## I. The Precambrian Era (4.5 billion – 600 million years ago)

አማርኛ

## I. የፕሪካምብሪያን ዘመን (ከ4.5 ቢሊዮን – 600 ሚሊዮን ዓመት በፊት)

ENGLISH

- The **oldest and longest** era.
- Frequent orogenic movements (mountain building, often with volcanic eruptions), intensive volcanism, later denudation and the formation of folded mountains.
- First life forms appeared (amoeba, jellyfish).
- The **basement complex** (old crystalline rock) formed. It lies under all other rocks.
- Outcrops in Ethiopia: central/northern Tigray, Mettekel, Assossa, Abbay Gorge, southern Sidama, Omo, Bale and Borena.

አማርኛ

- ጥንታዊውና ረጅሙ ዘመን።
- ተደጋጋሚ ተራራ ፈጣሪ (orogenic) እንቅስቃሴዎች፣ ከፍተኛ የእሳተ ገሞራ እንቅስቃሴ፣ በኋላ መሸርሸር፣ እና የታጠፉ ተራሮች አፈጣጠር።
- የመጀመሪያዎቹ የሕይወት ዓይነቶች ታዩ (አሜባ፣ ጄሊፊሽ)።
- የመሠረት ዐለት **(basement complex)** ተፈጠረ። ከሌሎች ዐለቶች ሁሉ በታች ይገኛል።
- በኢትዮጵያ የሚታዩባቸው ቦታዎች፦ መካከለኛና ሰሜናዊ ትግራይ፣ መተከል፣ አሶሳ፣ የአባይ ሸለቆ፣ ደቡብ ሲዳማ፣ ኦሞ፣ ባሌና ቦረና።

ENGLISH

### II. The Paleozoic Era (600 – 250 million years ago)

አማርኛ

### II. የፓሊዮዞይክ ዘመን (ከ600 – 250 ሚሊዮን ዓመት በፊት)

ENGLISH

- Dominant process: **denudation followed by peneplanation** (land reduced to a nearly flat surface).
- No significant structural formation; a **gap in rock formation** in Ethiopia.
- Massive denudation left **inselbergs** (residual hills).
- Life: **invertebrates** predominated.

አማርኛ

- ዋና ሂደት፦ መሸርሸርና ከዚያም ወደ ሜዳነት መዋረድ **(peneplanation)**።
- ጉልህ የመዋቅር ለውጥ አልተከሰተም፤ በኢትዮጵያ የዐለት አፈጣጠር ክፍተት አለ።
- ከፍተኛ መሸርሸር ኢንሰልበርግ **(inselberg)** ኮረብታዎችን አስቀረ።
- ሕይወት፦ ጀርባ አጥንት የሌላቸው እንስሳት የበዙበት።

ENGLISH

### III. The Mesozoic Era (250 – 70 million years ago)

አማርኛ

### III. የሜሶዞይክ ዘመን (ከ250 – 70 ሚሊዮን ዓመት በፊት)

ENGLISH

- Alternate slow sinking (**subsidence**) and uplift of the land: **epeirogenesis**. An era of **sedimentary rock** formation.
- **Triassic (250–180 Ma):** land sank in the south-east, the sea advanced north-westward (**transgression**). **Adigrat sandstone** formed: the oldest sedimentary rock, younger towards the north-west.
- **Jurassic (180–135 Ma):** transgression continued north-westward. **Hintalo limestone** formed (younger in the NW, older in the SE).
- **Cretaceous (135–70 Ma):** land began to rise; the sea **regressed** (retreated). **Upper sandstone** formed: the youngest sedimentary rock, lying on top of the others.
- These rocks were later buried under Cenozoic lava but are exposed in the south-eastern lowlands, central Tigray and the Abbay and Wabishebelle gorges.
- Life: reptiles (dinosaurs) predominated; at the end dinosaurs vanished and mammals, birds and flowering plants appeared.

አማርኛ

- የመሬት ዝግተኛ መውረድ **(subsidence)** እና መውጣት ተፈራርቀዋል፦ **epeirogenesis**። የደለል ዐለት **(sedimentary)** አፈጣጠር ዘመን።
- ትራይያሲክ **(250–180** ሚሊዮን ዓመት**)**፦ መሬት በደቡብ ምስራቅ ሰጠመ፤ ባሕር ወደ ሰሜን ምዕራብ ገሰገሰ **(transgression)**። የአድግራት አሸዋ ድንጋይ ተፈጠረ፤ ጥንታዊው ደለል ዐለት ሲሆን ወደ ሰሜን ምዕራብ እየወጣ ወጣት ነው።
- ጁራሲክ **(180–135** ሚሊዮን ዓመት**)**፦ የባሕር መስፋፋት ወደ ሰሜን ምዕራብ ቀጠለ። የሂንታሎ ኖራ ድንጋይ ተፈጠረ (በሰሜን ምዕራብ ወጣት፣ በደቡብ ምስራቅ ጥንታዊ)።

- ክሪቴሸስ **(135–70 ሚሊዮን ዓመት)**፦ መሬት መውጣት ጀመረ፤ ባሕር ሸሸ **(regression)**። የላይኛው አሸዋ ድንጋይ ተፈጠረ፤ በጣም ወጣቱ ደለል ዐለት ሲሆን በሌሎቹ ላይ ይገኛል።
- እነዚህ ዐለቶች በኋላ በሴኖዞይክ ላቫ ተሸፈኑ፤ ነገር ግን በደቡብ ምስራቅ ቆላዎች፣ በመካከለኛ ትግራይና በአባይና ዋቢ ሸበሌ ሸለቆዎች ይታያሉ።
- ሕይወት፦ ተሳቢዎች (ዳይኖሰሮች) የበዙበት፤ በመጨረሻ ዳይኖሰሮች ጠፉ፤ አጥቢ እንስሳት፣ አእዋፍና አበባ ያላቸው ተክሎች ታዩ።

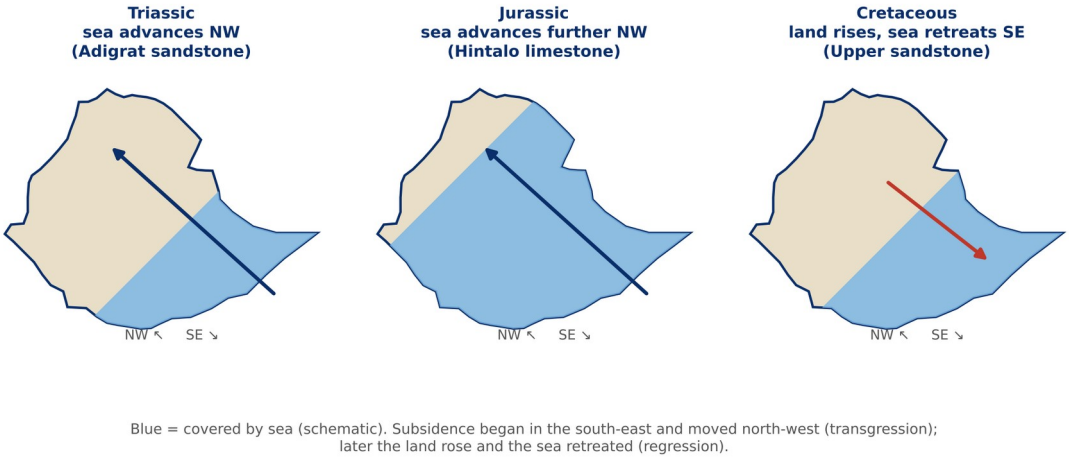



**Figure 1.10** *Mesozoic sea advance and retreat (schematic).*
ስዕል 1.10፦ የሚሶዞይክ የባሕር መስፋፋትና መሸሽ (ንድፍ)።

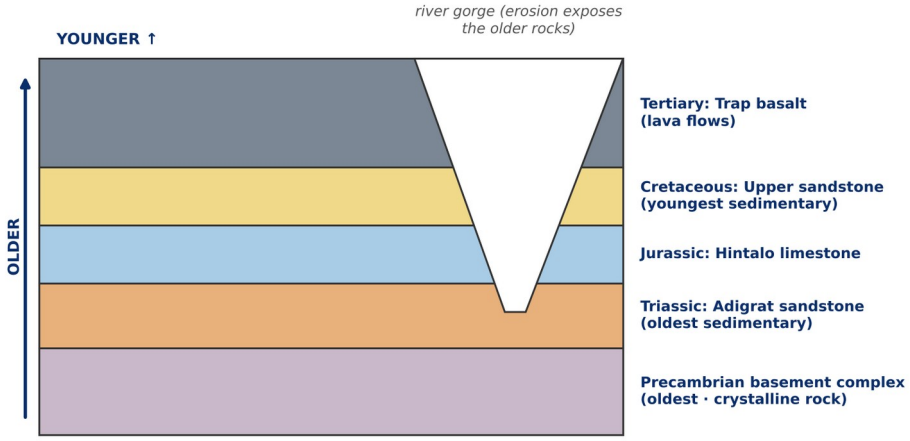



**Figure 1.11** *Order of rock layers in Ethiopia (oldest at the bottom). A river gorge exposes the older layers.*
ስዕል 1.11፦ የዐለት ንብርብሮች ቅደም ተከተል (ጥንታዊው ከታች)።

ENGLISH

## IV. The Cenozoic Era (70 million years ago – present)

አማርኛ

## IV. የሴኖዞይክ ዘመን (ከ70 ሚሊዮን ዓመት በፊት – አሁን)

ENGLISH

The most recent era, marked by major structural, climatic and biological events that shaped today's landforms. It has two periods: **Tertiary** and **Quaternary**.

### Tertiary (70 – 2 million years ago)

- **Uplift** continued to its maximum, forming huge dome-like structures.

- **Great cracks** opened in the crust and **basaltic lava** poured out (Trappean lava series).
- **Highlands and plateaus** formed: Northwestern Highlands, Southeastern Highlands and the Somali plateaus.
- Cracking and faulting formed the **Great East African Rift Valley System**, of which the Ethiopian Rift is a part.

### Quaternary (2 million years ago – present)

- Recent volcanism after the Rift Valley formed.
- Features: Afar Horst (extending into Djibouti); **Ertale** (active volcano, Afar); **Fentale** (dormant volcano, eastern Oromia); lava fields (Metahara); thermal springs and fumaroles; pluvial rainfall and deposition in the lowlands.
- The time of **modern human evolution**.

አማርኛ

> ከሁሉም የቅርብ ዘመን ሲሆን የዛሬዎቹን የመሬት ቅርጾች የቀረጹ ታላላቅ የመዋቅር፣ የአየር ንብረትና የሕይወት ክስተቶች ያሉበት ነው። ሁለት ክፍለ ዘመናት አሉት፦ ተርሸሪ እና ኳተርነሪ።

### ተርሸሪ (ከ70 – 2 ሚሊዮን ዓመት በፊት)

- ከፍታ **(uplift)** እስከ ከፍተኛው ቀጠለ፤ ግዙፍ የጉልላት መሰል መዋቅሮች ተፈጠሩ።
- በመሬት ቅርፊት ታላላቅ ስንጥቆች ተከፈቱ፤ የባሳልት ላቫ ፈሰሰ (Trappean lava series)።
- ደጋዎችና አምባዎች ተፈጠሩ፦ የሰሜን ምዕራብ ደጋ፣ የደቡብ ምስራቅ ደጋና የሶማሌ አምባዎች።
- ስንጥቅና ጥፋት (faulting) ታላቁን የምስራቅ አፍሪካ ስምጥ ሸለቆ ሥርዓት ፈጠሩ፤ የኢትዮጵያ ስምጥ የዚሁ አካል ነው።

### ኳተርነሪ (ከ2 ሚሊዮን ዓመት በፊት – አሁን)

- ስምጥ ሸለቆው ከተፈጠረ በኋላ የቅርብ ጊዜ የእሳተ ገሞራ እንቅስቃሴ።
- ገጽታዎች፦ የአፋር ሆርስት (እስከ ጅቡቲ የሚዘልቅ)፤ ኤርታሌ (ንቁ እሳተ ገሞራ፣ አፋር)፤ ፈንታሌ (የተኛ እሳተ ገሞራ፣ ምስራቅ ኦሮሚያ)፤ የላቫ ሜዳዎች (መተሃራ)፤ ፍል ውሃና የእንፋሎት ቀዳዳዎች፤ የዝናብ ዘመን ዝናብና በቆላዎች ላይ ደለል መጣል።
- የዘመናዊ ሰው ዝግመተ ለውጥ ዘመን።

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**COMMON MISCONCEPTIONS** የተለመዱ የተሳሳቱ ግንዛቤዎች

× The Mesozoic is the era of basalt lava and the Rift Valley.
✓ Lava (Trap basalt) and the Rift Valley belong to the CENOZOIC (Tertiary). The Mesozoic made SEDIMENTARY rocks.
ባሳልት ላቫና ስምጥ ሸለቆ የሴኖዞይክ ናቸው፤ ሜሶዞይክ ደለል ዐለቶችን ፈጠረ።

× The Cretaceous sandstone is the oldest sedimentary rock.
✓ The OLDEST is Triassic Adigrat sandstone; the Upper (Cretaceous) sandstone is the YOUNGEST.
ጥንታዊው የትራይያሲክ የአድግራት አሸዋ ድንጋይ ነው፤ የላይኛው (ክሪቴሸስ) ወጣቱ ነው።

× The sea advanced from the north-west.
✓ Subsidence began in the SOUTH-EAST and the sea advanced NORTH-WESTWARD; later the sea retreated south-east.
ባሕሩ ከደቡብ ምስራቅ ወደ ሰሜን ምዕራብ ገሰገሰ።

× Ertale is dormant and Fentale is active.
✓ ERTALE = active (Afar); FENTALE = dormant (eastern Oromia).
ኤርታሌ ንቁ፤ ፈንታሌ የተኛ ነው።

× The Paleozoic is the era of dinosaurs.
✓ Paleozoic = invertebrates; Mesozoic = reptiles/dinosaurs.
ፓሊዮዞይክ ጀርባ አጥንት የሌላቸው፤ ሜሶዞይክ ተሳቢዎች።

## Worked Example 1.10 · Order the rocks

### ምሳሌ 1.10፡ ዐለቶችን በቅደም ተከተል ማስቀመጥ

**Question:** Arrange from oldest to youngest: Hintalo limestone, Trap basalt, Basement complex, Upper sandstone, Adigrat sandstone.

**Answer:** Basement complex (Precambrian) → Adigrat sandstone (Triassic) → Hintalo limestone (Jurassic) → Upper sandstone (Cretaceous) → Trap basalt (Tertiary).

> መልስ፦ የመሠረት ዐለት (ፕሪካምብሪያን) → አድግራት አሸዋ ድንጋይ (ትራይያሲክ) → ሂንታሎ ኖራ ድንጋይ (ጁራሲክ) → የላይኛው አሸዋ ድንጋይ (ክሪቴሸስ) → ትራፕ ባሳልት (ተርሸሪ)።

## Worked Example 1.11 · Why do sandstone and limestone outcrop in Tigray?

### ምሳሌ 1.11፡ በትግራይ አሸዋና ኖራ ድንጋይ ለምን ይታያሉ?

**Answer:** The Mesozoic sedimentary rocks were later covered by Tertiary basalt. Over a very long time erosion removed the basalt in places such as the Plateau of Tigray and the river gorges, exposing the older sedimentary rocks beneath.

> መልስ፦ የሜሶዞይክ ደለል ዐለቶች በኋላ በተርሸሪ ባሳልት ተሸፍነው ነበር። ረጅም ጊዜ በፊጀ መሸርሸር ባሳልቱ እንደ ትግራይ አምባና የወንዝ ሸለቆዎች ባሉ ቦታዎች ተወግዶ ከስር ያሉ ደለል ዐለቶች ተገለጡ።

### Worked Example 1.12 · Match the period to the event

**ምሳሌ 1.12፡ ክፍለ ዘመኑን ከክስተቱ ጋር አዛምድ**

**Question:** In which era/period did each happen? (a) Adigrat sandstone formed (b) Great cracks and trap lava (c) Peneplanation (d) Ertale-type volcanism (e) Sea retreat.

**Answer:** (a) Mesozoic – Triassic; (b) Cenozoic – Tertiary; (c) Paleozoic; (d) Cenozoic – Quaternary; (e) Mesozoic – Cretaceous.

> መልስ፦ (ሀ) ሜሶዞይክ – ትራይያሲክ፤ (ለ) ሴኖዞይክ – ተርሸሪ፤ (ሐ) ፓሊዮዞይክ፤ (መ) ሴኖዞይክ – ኳተርነሪ፤ (ሠ) ሜሶዞይክ – ክሪቴሸስ።

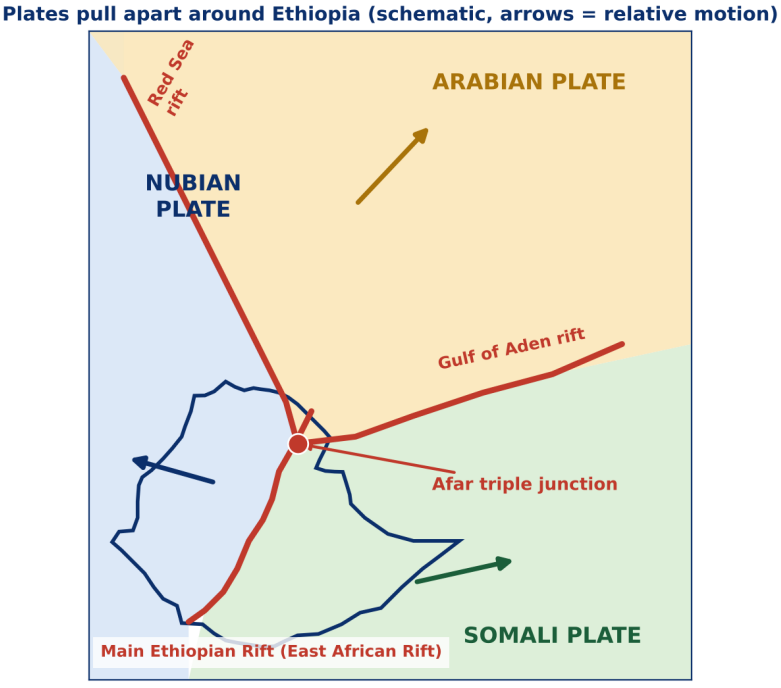



**Figure 1.12** *Plates around Ethiopia and the Afar triple junction (schematic).*
ስዕል 1.12፦ በኢትዮጵያ ዙሪያ ያሉ ሰሌዳዎችና የአፋር ሦስትዮሽ መገናኛ (ንድፍ)።

*Enrichment (beyond the notes): the plate-tectonic picture above explains WHY the crust cracks in the Cenozoic; the exam answers should use the terms in the notes.*

**KEY TAKEAWAYS** **Geological history / የጂኦሎጂ ታሪክ**

- Eras: Precambrian (4.5 bn–600 Ma), Paleozoic (600–250), Mesozoic (250–70), Cenozoic (70–0).
  አራት ዘመናት።
- Precambrian = basement complex; Paleozoic = denudation/peneplanation; Mesozoic = sedimentary rocks; Cenozoic = lava, highlands, Rift, volcanoes.
  ፕሪካምብሪያን = የመሠረት ዐለት፤ ፓሊዮዞይክ = መሸርሸር፤ ሜሶዞይክ = ደለል፤ ሴኖዞይክ = ላቫና ስምጥ።
- Mesozoic order: Triassic (Adigrat) → Jurassic (Hintalo) → Cretaceous (Upper sandstone).
  ትራይያሲክ → ጁራሲክ → ክሪቴሽስ።
- Tertiary: uplift, cracks, trap basalt, highlands, Rift. Quaternary: Ertale, Fentale, Metahara, hot springs, human evolution.
  ተርሸሪ፣ ኳተርነሪ።

## 1.3.2 Landforms of Ethiopia

## 1.3.2 የኢትዮጵያ የመሬት ቅርጾች

### ENGLISH

Landforms are individual features of the earth's surface. In Ethiopia they are mostly the result of **Cenozoic** tectonic and volcanic activity. They include highlands, plateaus, **ambas** (flat-topped mountains), rugged mountains, deep gorges and lowlands.

- **Highest point:** Ras Dashen, **4,620 m** above mean sea level.
- **Lowest point:** Dallol depression, **−116 m** (116 m below sea level).
- **Relief range:** 4,620 − (−116) = **4,736 m**.
- **56 %** of the country is highland (above **1,000 m**). Ethiopia is called the **"Roof of Eastern Africa"**.
- Two opposing forces shape landforms: **endogenic** (inside the Earth: volcanism, tectonics) and **exogenic** (at the surface: denudation, peneplanation).

### አማርኛ

> የመሬት ቅርጾች የምድር ገጽ ነጠላ ገጽታዎች ናቸው። በኢትዮጵያ በአብዛኛው የሴኖዞይክ የቴክቶኒክና የእሳተ ገሞራ እንቅስቃሴ ውጤት ናቸው። ደጋዎች፣ አምባዎች፣ አምባ **(amba)** የሚባሉ ጠፍጣፋ አናት ያላቸው ተራሮች፣ ዓለታማ ተራሮች፣ ጥልቅ ሸለቆዎችና ቆላዎች ይገኙባቸዋል።

- ከፍተኛው ነጥብ፦ ራስ ዳሸን፣ ከባሕር ጠለል በላይ **4,620** ሜ።
- ዝቅተኛው ነጥብ፦ የዳሎል ዝቅታ፣ **−116** ሜ (ከባሕር ጠለል በታች 116 ሜ)።
- የከፍታ ልዩነት፦ 4,620 − (−116) = **4,736** ሜ።
- የአገሪቱ **56 %** ደጋ ነው (ከ**1,000** ሜ በላይ)። ኢትዮጵያ "የምስራቅ አፍሪካ ጣሪያ" ትባላለች።
- ሁለት ተቃራኒ ኃይሎች የመሬት ቅርጾችን ይቀርጻሉ፦ የውስጥ **(endogenic)** (እሳተ ገሞራ፣ ቴክቶኒክ) እና የውጭ **(exogenic)** (መሸርሸር፣ ወደ ሚዳነት መዋረድ)።

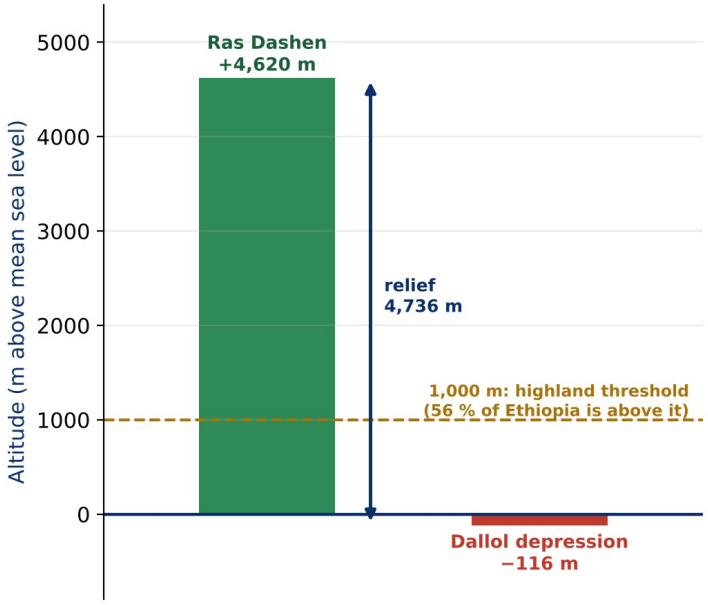



**Figure 1.13** *Highest and lowest points of Ethiopia.*
ስዕል 1.13፦ የኢትዮጵያ ከፍተኛና ዝቅተኛ ነጥቦች።

## Worked Example 1.13 · Relief and highland area

### ምሳሌ 1.13፡ የከፍታ ልዩነትና የደጋ ስፋት

**Question:** (a) Find the relief (difference between highest and lowest points). (b) Estimate the highland area if 56 % of 1,106,000 km² lies above 1,000 m.

(a) $4{,}620 - (-116) = 4{,}620 + 116 = 4{,}736$ m

(b) $0.56 \times 1{,}106{,}000 = 619{,}360$ km²

መልስ፦ (ሀ) 4,620 + 116 = 4,736 ሜ። (ለ) 0.56 × 1,106,000 = 619,360 ካ.ኪ.ሜ ደጋ።

## COMMON MISCONCEPTIONS የተለመዱ የተሳሳቱ ግንዛቤዎች

× The Dallol depression is 116 m above sea level.
✓ It is 116 m BELOW sea level (−116 m).
ዳሎል ከባሕር ጠለል በታች 116 ሜ ነው።

× Endogenic forces include erosion.
✓ Erosion/denudation is EXOGENIC. Endogenic = volcanic and tectonic forces from inside the Earth.
መሸርሸር የውጭ ኃይል ነው።

× Highland means above 500 m.
✓ In the notes, highland = above 1,000 m (56 % of Ethiopia). The lowlands are mostly 500–1,000 m.
ደጋ ማለት ከ1,000 ሜ በላይ ነው።

## Three Physiographic Divisions

### ሦስት የመሬት አቀማመጥ ክፍሎች

ENGLISH

Ethiopian relief is divided into three main physiographic divisions:

- **I. The Western Highlands and associated lowlands** (44 % of the country, the largest).
- **II. The Southeastern Highlands and associated lowlands.**
- **III. The Rift Valley** (18 % of the country), which separates the two highland blocks.

አማርኛ

> የኢትዮጵያ የመሬት አቀማመጥ በሦስት ዋና ክፍሎች ይከፈላል፦

- **I.** የምዕራብ ደጋና ተያያዥ ቆላዎች (የአገሪቱ 44 %፣ ትልቁ)።
- **II.** የደቡብ ምስራቅ ደጋና ተያያዥ ቆላዎች።
- **III.** ስምጥ ሸለቆ (የአገሪቱ 18 %)፤ ሁለቱን የደጋ ክፍሎች ይለያል።

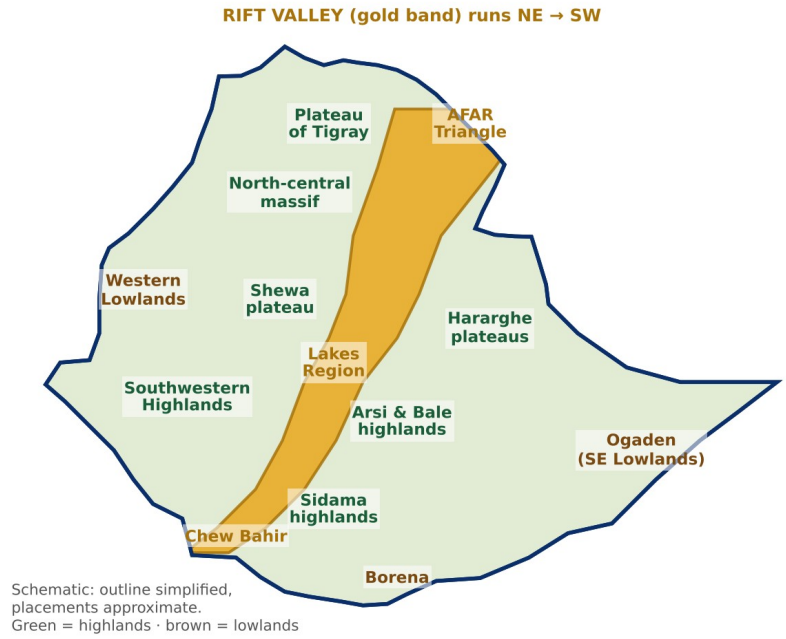



**Figure 1.14** *The physiographic divisions of Ethiopia (schematic).*
ስዕል 1.14፦ የኢትዮጵያ የመሬት አቀማመጥ ክፍሎች (ንድፍ)።

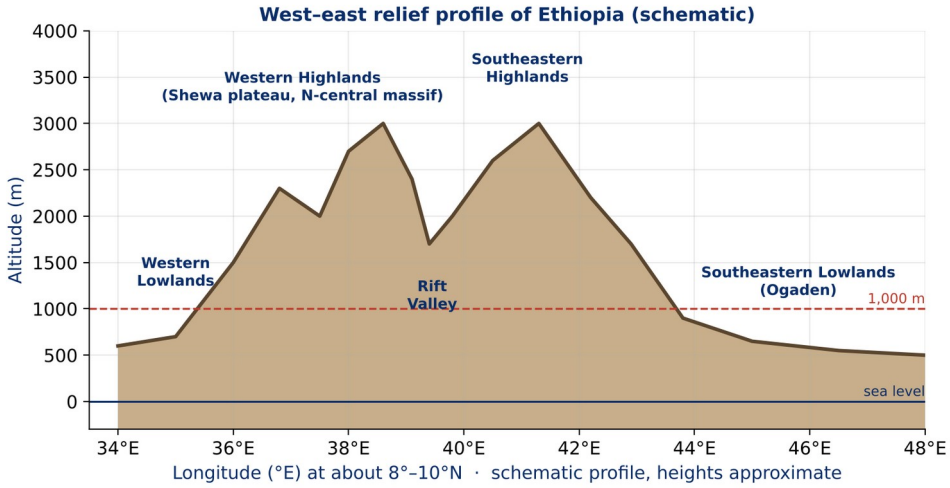



**Figure 1.15** *West–east relief profile (schematic).*
ስዕል 1.15፦ የምዕራብ–ምስራቅ የከፍታ መስቀለኛ ክፍል (ንድፍ)።

ENGLISH

## I-A. The Western Highlands

አማርኛ

## I-ሀ. የምዕራብ ደጋዎች

ENGLISH

Stretch from Tigray in the north to the Gamo–Goffa highlands in the south-west. They are the source of the **Abbay, Baro and Tekezze** rivers, which matters for geopolitics. Four sub-divisions:

አማርኛ

> ከሰሜን ትግራይ እስከ ደቡብ ምዕራብ የጋሞ–ጎፋ ደጋዎች ይዘልቃሉ። የአባይ፣ ባሮና ተከዜ ወንዞች ምንጭ ናቸው፤ ይህም ለጂኦፖለቲካ ጠቃሚ ነው። አራት ንዑስ ክፍሎች አሉ፦

| Sub-division<br>ንዑስ ክፍል | Key facts | Peaks, rivers & farming |
|---|---|---|
| **Plateau of Tigray** | Northernmost; between Mereb R. (N) and Tekezze R. (S); eroded sandstone and limestone, thin poor soils, severe erosion; peaks over 3,000 m | Tsibet 3,988 m · Ambalage 3,291 m · Assimba 3,248 m |
| **North-Central Massif** | Between Abbay Gorge (S) and Tekezze Gorge (N); Gondar, Gojjam, Wollo; most rugged and dissected; capped by Tertiary basalt; ambas; Yeju–Wadla Delanta land bridge links Gondar/Gojjam to Wollo; massifs: South Gondar (Semien), Lasta & Wollo, Gojjam (most extensive tableland) | Ras Dashen 4,620 · Legeda 4,532 · Analu 4,480 · Tefaw Lezer 4,456 · Kolo 4,300 · Guna 4,231 · Abuna Yoseph 4,190 · Hey 4,154. Cereals (teff), pulses, oil seeds |
| **Plateau of Shewa** | Smallest sub-division; dome-shaped; watershed of Awash, Omo-Gibe and Abbay basins; crescent shape makes the Abbay swerve north | Abbuye Meda 4,000 · Guraghe 3,721. Abbay (W) and Awash (E) tributaries |

| Sub-division<br>ንዑስ ክፍል | Key facts | Peaks, rivers & farming |
|---|---|---|
| **Southwestern Highlands** | South of the Abbay trough; wettest region (>1,500 mm/yr); Wollega, Illubabor, Jimma, Kafa, Gamo, Goffa; few areas above 2,500 m | Mt Gughe 4,200 (highest here). Rivers: Dabus & Didesa (N), Baro-Akobo (W), Omo-Gibe (S), Gojeb-Gibe (E). Coffee, spices, timber, rubber, cattle, honey, maize |

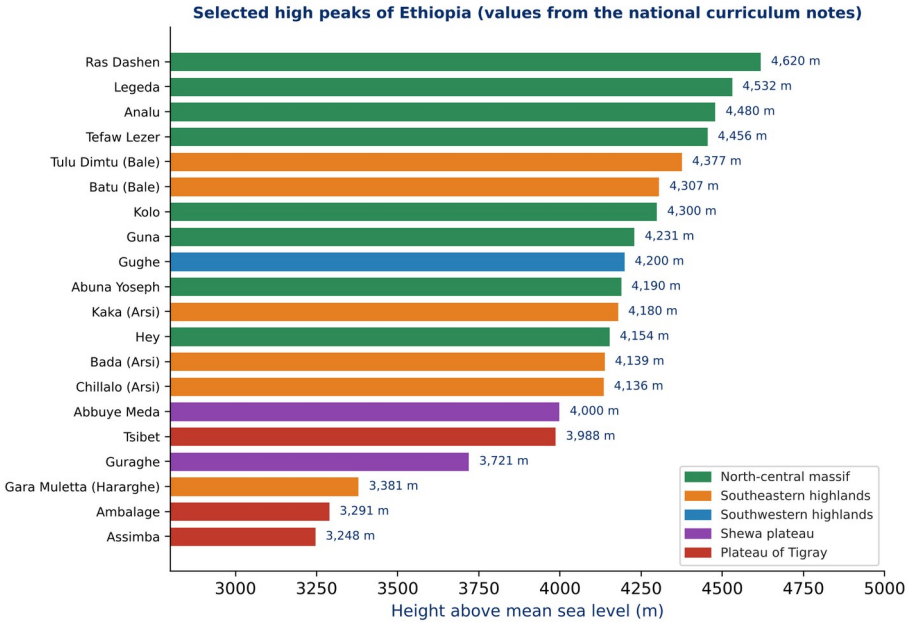



**Figure 1.16** *Selected high peaks of Ethiopia (values from the study notes).*
ስዕል 1.16፦ የተመረጡ የኢትዮጵያ ከፍተኛ ጫፎች (ከጥናት ማስታወሻው የተወሰዱ)።

ENGLISH

## I-B. The Western Lowlands

አማርኛ

## I-ለ. የምዕራብ ቆላዎች

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ENGLISH

- Extent: western Tigray to southern Gamo–Goffa, along the Sudan and South Sudan border. Altitude **500–1,000 m**.
- Climate: arid and semi-arid; the **Baro-Akobo lowland is the wettest** (Baro River).
- Sub-divisions: Tekezze–Angereb, Abbay–Dinder, Baro-Akobo and Omo-Gibe lowlands.
- People: pastoralist and semi-pastoralist. Border business towns: Humera, Kurmuk, Omedla, Metema.

አማርኛ

- ስፋት፦ ከምዕራብ ትግራይ እስከ ደቡብ ጋሞ–ጎፋ፣ ከሱዳንና ደቡብ ሱዳን ድንበር ጋር። ከፍታ **500–1,000** ሜ።
- የአየር ንብረት፦ ደረቅና ከፊል ደረቅ፤ ባሮ-አኮቦ ቆላ እርጥብ ነው (የባሮ ወንዝ)።
- ንዑስ ክፍሎች፦ ተከዜ–አንገረብ፣ አባይ–ዲንደር፣ ባሮ-አኮቦና ኦሞ-ጊቤ ቆላዎች።

- ነዋሪዎች፦ አርብቶ አደርና ከፊል አርብቶ አደር። የድንበር ንግድ ከተሞች፦ ሁመራ፣ ኩርሙክ፣ ኦሚድላ፣ መተማ።

ENGLISH

## II-A. The Southeastern Highlands

አማርኛ

## II-ሀ. የደቡብ ምስራቅ ደጋዎች

ENGLISH

Southeast of the Rift Valley; capped by basalt like the other highlands. They are the main source of the **Wabe Shebelle and Genale** rivers. Bounded by the Rift fault line (W), Ogaden Lowlands (E) and the Elkerie and Borena Lowlands (S).

አማርኛ

> ከስምጥ ሸለቆ በደቡብ ምስራቅ ይገኛሉ፤ እንደሌሎቹ ደጋዎች በባሳልት የተሸፈኑ ናቸው። የዋቢ ሸበሌና ገናሌ ወንዞች ዋና ምንጭ ናቸው። በስምጥ ስንጥቅ መስመር (ምዕራብ)፣ በኦጋዴን ቆላ (ምስራቅ)፣ በኤልከሬና በቦረና ቆላዎች (ደቡብ) ይከበባሉ።

| Sub-division<br>ንዑስ ክፍል | Key facts | Peaks & farming |
|---|---|---|
| **Hararghe plateaus** | Rise sharply from the Rift fault line, extend gently east to Jigjiga, then fall to the Ogaden; left-bank tributaries of the Wabe Shebelle; basalt eroded exposing limestone | Gara Muletta 3,381 · Jebel Tita 3,122. Coffee, chat, sorghum, millet |
| **Arsi plateau** | Rolling plateau good for farming; Gugu and Chillalo massifs | Chillalo 4,136 · Bada 4,139 · Kaka 4,180. Wheat |
| **Bale Massif** | Next to Arsi, separated by Wabe Shebelle headstreams; north flat basalt plateau; south huge mountains | Tulu Dimtu 4,377 · Batu 4,307. Barley |
| **Sidama highlands** | SW extension; separated from Bale by the Genale valley; slopes gently south and south-east; Genale drainage | — |

ENGLISH

## II-B. The Southeastern Lowlands

አማርኛ

## II-ለ. የደቡብ ምስራቅ ቆላዎች

ENGLISH

- Foothills along the Ethio-Somalia and Ethio-Kenya borders; **500–1,000 m**.
- Plains: **Ogaden, Elkeri, Borena and Genale**.
- Climate: extreme, with rainfall often **below 500 mm**.
- Geology: sandstones and recent marine deposits. People: pastoralists.

አማርኛ

- ከኢትዮጵያ–ሶማሊያና ኢትዮጵያ–ኬንያ ድንበሮች ጋር ያሉ ኮረብታማ ግርጌዎች፤ **500–1,000** ሜ።

- ሜዳዎች፦ ኦጋዴን፣ ኤልከሬ፣ ቦረናና ገናሌ።
- የአየር ንብረት፦ ጽንፈኛ፤ ዝናብ ብዙ ጊዜ ከ**500** ሚ.ሜ በታች።
- ጂኦሎጂ፦ አሸዋ ድንጋዮችና የቅርብ ጊዜ የባሕር ደለሎች። ነዋሪዎች፦ አርብቶ አደሮች።

ENGLISH

## III. The Rift Valley

አማርኛ

## III. ስምጥ ሸለቆ

ENGLISH

A narrow strip of land formed by fractures in the crust. The Ethiopian Rift is part of the **Great East African Rift**, from the Afar Depression in the north, through the Rift Valley Lakes region, to Lake Turkana.

- Formed at the end of the **Tertiary** by tectonic (epeirogenic) activity.
- Evidence of instability: active volcanoes, minor faulting, hot springs, cinder cones, fumaroles, geysers.
- Runs **north-east to south-west** and divides the highlands.
- Length **1,700 km** (Ethiopia and Eritrea); **18 %** of Ethiopia's area.

አማርኛ

> በመሬት ቅርፊት ስንጥቅ የተፈጠረ ጠባብ የመሬት ንጣፍ ነው። የኢትዮጵያ ስምጥ ከሰሜን የአፋር ዝቅታ፣ በስምጥ ሸለቆ ሐይቆች አካባቢ አልፎ እስከ ቱርካና ሐይቅ የሚዘልቀው የታላቁ የምስራቅ አፍሪካ ስምጥ አካል ነው።

- በተርሸሪ መጨረሻ በቴክቶኒክ (epeirogenic) እንቅስቃሴ ተፈጠረ።
- አለመረጋጋት ማስረጃዎች፦ ንቁ እሳተ ገሞራዎች፣ ጥቃቅን ጥፋቶች፣ ፍል ውሃዎች፣ የአመድ ኮኖች፣ የእንፋሎት ቀዳዳዎች፣ ጋይዘሮች።
- ከሰሜን ምስራቅ ወደ ደቡብ ምዕራብ ይሄዳል፤ ደጋዎችን ይከፍላል።
- ርዝመት **1,700** ኪ.ሜ (ኢትዮጵያና ኤርትራ)፤ የኢትዮጵያ **18 %**።

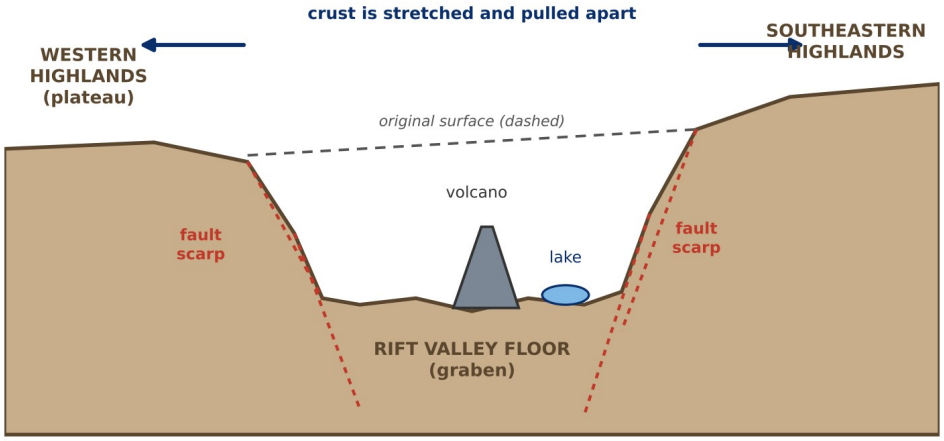



**Figure 1.17** *Cross-section of the Rift Valley (schematic).*
ስዕል 1.17፦ የስምጥ ሸለቆ መስቀለኛ ክፍል (ንድፍ)።

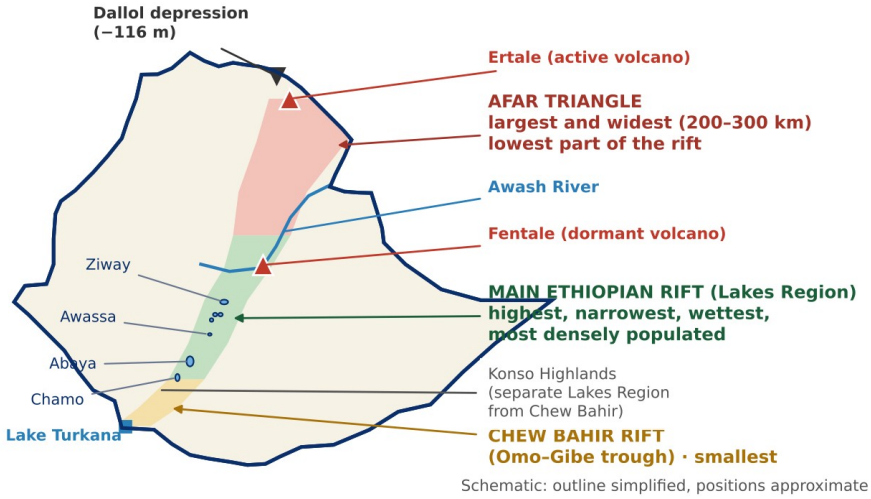



**Figure 1.18** *The three parts of the Ethiopian Rift (schematic map).*
ስዕል 1.18፦ የኢትዮጵያ ስምጥ ሥስት ክፍሎች (ንድፍ ካርታ)።

| Part<br>ክፍል | Position | Key facts |
|---|---|---|
| **Afar Triangle** | North | Largest and widest (200–300 km); lowest (−116 m at Kobar Sink to 900 m at Awash); Dallol depression, grabens, cinder cones, salt plains and lakes (Assale, Afrera); Mt Fentale; fossil-rich, ancient hominid sites |
| **Main Ethiopian Rift (Lakes Region)** | Centre | Lower Awash basin to Lake Chamo; most elevated and narrowest; wettest, most vegetated, most densely populated; sedentary farming; many lakes |
| **Chew Bahir Rift (Omo–Gibe trough)** | South | Smallest part; separated from the Lakes Region by the Konso Highlands; shallow marshes of tall grass where Sagan and Woito end; split by the Amaro range near Arba Minch into Ganjuli and Galena valleys; pastoralists |

**COMMON MISCONCEPTIONS** የተለመዱ የተሳሳቱ ግንዛቤዎች

× The Afar Triangle is the narrowest part of the Rift.
✓ It is the LARGEST and WIDEST (200–300 km) and the lowest. The Lakes Region is the narrowest and highest.
የአፋር ሦስት ማዕዘን ትልቁና ሰፊው ነው፤ ጠባቡ የሐይቆች አካባቢ ነው።

× The Rift Valley divides Ethiopia from north to south.
✓ It runs NORTH-EAST to SOUTH-WEST.
ስምጥ ከሰሜን ምስራቅ ወደ ደቡብ ምዕራብ ይሄዳል።

× The western highlands are the smallest physiographic division.
✓ They are the LARGEST (44 %). The Shewa plateau is the smallest sub-division of the western highlands.
የምዕራብ ደጋ ትልቁ ነው (44 %)።

× The Southwestern Highlands are the highest-rainfall AND highest-altitude highlands.
✓ They are the wettest (>1,500 mm) but generally LOWER than the northern/eastern highlands.
የደቡብ ምዕራብ ደጋ እርጥብ ነው ግን ከሰሜንና ምስራቅ ደጋዎች ዝቅ ይላል።

### Worked Example 1.14 · Areas of the divisions

**ምሳሌ 1.14፡ የክፍሎች ስፋት**

**Question:** Find the approximate area of (a) the Western Highlands (44 %) and (b) the Rift Valley (18 %). What share is left for the Southeastern division?

(a) $0.44 \times 1{,}106{,}000 = 486{,}640 \text{ km}^2$

(b) $0.18 \times 1{,}106{,}000 = 199{,}080 \text{ km}^2$

(c) $\text{Remainder} = 100 - 44 - 18 = 38\ \% \ \rightarrow\ 0.38 \times 1{,}106{,}000 = 420{,}280 \text{ km}^2$

> መልስ፦ (ሀ) 486,640 ካ.ኪ.ሜ፤ (ለ) 199,080 ካ.ኪ.ሜ፤ (ሐ) ለደቡብ ምስራቅ ክፍል የቀረው 38 % ≈ 420,280 ካ.ኪ.ሜ (የቀረው ስሌት ከመማሪያ ማስታወሻው መቶኛዎች የተገኘ ግምት ነው)።

### Worked Example 1.15 · Which region?

**ምሳሌ 1.15፡ የትኛው ክልል?**

**Question:** Name the region for each: (a) source of the Wabe Shebelle and Genale; (b) wettest lowland; (c) watershed of Awash, Omo-Gibe and Abbay; (d) wheat on rolling plateau; (e) narrowest, most populated part of the Rift.

**Answer:** (a) Southeastern Highlands; (b) Baro-Akobo lowland; (c) Plateau of Shewa; (d) Arsi plateau; (e) Main Ethiopian Rift (Lakes Region).

> መልስ፦ (ሀ) የደቡብ ምስራቅ ደጋ፤ (ለ) የባሮ-አኮቦ ቆላ፤ (ሐ) የሸዋ አምባ፤ (መ) የአርሲ አምባ፤ (ሠ) ዋናው የኢትዮጵያ ስምጥ (የሐይቆች አካባቢ)።

### KEY TAKEAWAYS Landforms / የመሬት ቅርጾች

- Highest Ras Dashen 4,620 m; lowest Dallol −116 m; 56 % highland (>1,000 m); “Roof of Eastern Africa”.
  ከፍተኛ፣ ዝቅተኛ፣ 56 % ደጋ።
- Divisions: Western (44 %), Southeastern, Rift Valley (18 %, 1,700 km, NE→SW).
  ሦስት ክፍሎች።
- Rift parts: Afar Triangle (largest, lowest) · Main Ethiopian Rift (highest, narrowest, most populated) · Chew Bahir (smallest).
  ሦስት የስምጥ ክፍሎች።
- Wettest highland: Southwestern (>1,500 mm). Wettest lowland: Baro-Akobo.
  እርጥብ ደጋና ቆላ።

# Unit Summary and Formula Sheet

## የምዕራፉ ማጠቃለያና የቀመር ሉህ

### ENGLISH

Ethiopia lies in Eastern Africa between 3°N–15°N and 33°E–48°E, is landlocked, and has a generally compact shape. It is the tenth largest country in Africa (1,106,000 km²). Its geology is the product of four eras; its landforms are mostly Cenozoic in origin and range from −116 m (Dallol) to 4,620 m (Ras Dashen).

### አማርኛ

> ኢትዮጵያ በምስራቅ አፍሪካ ከ3°N–15°N እና 33°E–48°E መካከል ትገኛለች፤ የባሕር በር የላትም፤ በአጠቃላይ ውሱን ቅርጽ አላት። በአፍሪካ አሥረኛዋ ትልቅ አገር ናት (1,106,000 ካ.ኪ.ሜ)። ጂኦሎጂዋ የአራት ዘመናት ውጤት ነው፤ የመሬት ቅርጾቿ በአብዛኛው የሴኖዞይክ ሲሆኑ ከ−116 ሜ (ዳሎል) እስከ 4,620 ሜ (ራስ ዳሸን) ይደርሳሉ።

## Key numbers to memorise

### በቃል የሚያዙ ቁጥሮች

| Fact | Value | አማርኛ |
|---|---|---|
| **Latitude / longitude** | 3°N–15°N / 33°E–48°E | ኬክሮስ / ኬንትሮስ |
| **Area** | 1,106,000 km² (10th in Africa) | ስፋት |
| **Share of world land / Africa** | 0.7 % / 3.6 % | ድርሻ |
| **Boundary (longest / shortest)** | 5,260 km (Somalia 1,600 / Djibouti 310) | ድንበር |
| **B/C ratio** | 1.411 (41 % above a circle) | B/C |
| **A/B ratio** | ≈ 210 km² per km | A/B |
| **Highest / lowest point** | Ras Dashen 4,620 m / Dallol −116 m | ከፍተኛ / ዝቅተኛ |
| **Highland share (>1,000 m)** | 56 % | ደጋ |
| **Western Highlands / Rift Valley share** | 44 % / 18 % | ድርሻ |
| **Rift length** | 1,700 km (Ethiopia + Eritrea) | ርዝመት |
| **Afar Triangle width** | 200–300 km | ስፋት |
| **Eras (years ago)** | 4.5 bn–600 M · 600–250 M · 250–70 M · 70 M–now | ዘመናት |
| **Triassic · Jurassic · Cretaceous** | 250–180 · 180–135 · 135–70 million | ክፍለ ዘመናት |

| Fact | Value | አማርኛ |
|---|---|---|
| **Tertiary · Quaternary** | 70–2 million · 2 million–now | ተርሸሪ · ኳተርነሪ |

## Formulas

ቀመሮች

$r = \sqrt{(A \div \pi)}$ $C = 2\pi r$ $B/C = B \div C$ (closer to 1 = more compact)
$A/B$ = Area ÷ Boundary (larger = more compact)
Relief = highest − lowest = 4,620 − (−116) = 4,736 m
Part area = percentage × 1,106,000 km² (e.g. 0.56 × 1,106,000 = 619,360)

## Era-by-era summary

ዘመን በዘመን ማጠቃለያ

| Era | What happened in Ethiopia | Rock / landform |
|---|---|---|
| **Precambrian** | Orogeny, volcanism, folded mountains; first life | Basement complex |
| **Paleozoic** | Denudation, peneplanation, no structures | Inselbergs; gap in rocks |
| **Mesozoic** | Subsidence/uplift, sea advance then retreat | Adigrat → Hintalo → Upper sandstone |
| **Cenozoic: Tertiary** | Uplift, cracking, lava, Rift formed | Trap basalt; highlands; plateaus |
| **Cenozoic: Quaternary** | Recent volcanism; human evolution | Ertale, Fentale, Metahara lava |

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# Exam Strategy and Memory Aids

## የፈተና ስልትና የማስታወሻ ዘዴዎች

### ENGLISH

#### Mnemonics

- **Eras (old → young): P-P-M-C** – "Precambrian, Paleozoic, Mesozoic, Cenozoic". Life: **amoeba → invertebrates → reptiles → mammals/humans**.
- **Mesozoic rocks: A-H-U** – **A**digrat (Triassic), **H**intalo (Jurassic), **U**pper sandstone (Cretaceous). Alphabetical order = age order.
- **Rift (north → south): A-M-C** – **A**far Triangle (largest, lowest), **M**ain Ethiopian Rift (highest, narrowest), **C**hew Bahir (smallest).
- **Vicinal = Visit neighbours; Strategic = Seas and landmasses; Absolute = Angles (lat/long).**
- **Ertale = Eruptive (active); Fentale = Fast asleep (dormant).**
- **Physical branches (5):** G-C-B-S-O = Geomorphology, Climatology, Biogeography, Soil, Oceanography. **Human (6):** P-C-E-P-U-H = Population, Cultural, Economic, Political, Urban, Historical.

#### Time-saving rules

- Read ALL four options. The exam often places a half-true option before the correct one.
- Words that signal traps: NOT, EXCEPT, INCORRECT, LARGEST/SMALLEST, OLDEST/YOUNGEST, ACTIVE/DORMANT.
- For calculations, write the formula first, then substitute, then check the units (km, km², m).
- Map/figure questions: find the compass direction first, then the neighbours or layers.

### አማርኛ

#### የማስታወሻ ዘዴዎች

- ዘመናት (ከጥንት → ወደ ወጣት)፦ ፕ-ፓ-ሜ-ሴ – ፕሪካምብሪያን፣ ፓሊዮዞይክ፣ ሜሶዞይክ፣ ሴኖዞይክ። ሕይወት፦ አሜባ → ጀርባ አጥንት የሌላቸው → ተሳቢዎች → አጥቢዎች/ሰው።
- የሜሶዞይክ ዐለቶች፦ **A-H-U** – አድግራት (ትራይያሲክ)፣ ሂንታሎ (ጁራሲክ)፣ የላይኛው አሸዋ (ክሪቴሸስ)። ቅደም ተከተሉ የዕድሜ ቅደም ተከተል ነው።
- ስምጥ (ሰሜን → ደቡብ)፦ አ-ዋ-ጨ – አፋር ሦስት ማዕዘን (ትልቅ፣ ዝቅተኛ)፣ ዋናው ስምጥ (ከፍተኛ፣ ጠባብ)፣ ጨው ባሕር (ትንሹ)።
- አጎራባች = ጎረቤቶች፤ ስትራቴጂያዊ = ባሕርና የየብስ ክፍሎች፤ ፍጹም = ኬክሮስና ኬንትሮስ።
- ኤርታሌ = ንቁ፤ ፈንታሌ = የተኛ።
- ተፈጥሯዊ ቅርንጫፎች (5)፦ ጂኦሞርፎሎጂ፣ ክሊማቶሎጂ፣ ባዮጂኦግራፊ፣ አፈር፣ ውቅያኖስ። ሰዋዊ (6)፦ ሕዝብ፣ ባህል፣ ኢኮኖሚ፣ ፖለቲካ፣ ከተማ፣ ታሪካዊ።

#### ጊዜ ቆጣቢ ደንቦች

- አራቱንም አማራጮች ያንብቡ። ፈተናው ብዙ ጊዜ ከትክክለኛው መልስ በፊት ግማሽ እውነት የሆነ አማራጭ ያስቀምጣል።
- የወጥመድ ቃላት፦ NOT፣ EXCEPT፣ INCORRECT፣ LARGEST/SMALLEST፣ OLDEST/YOUNGEST፣ ACTIVE/DORMANT።

- ለስሌቶች መጀመሪያ ቀመሩን ይጻፉ፣ ከዚያ ይተኩ፣ በመጨረሻ መለኪያውን (ኪ.ሜ፣ ካ.ኪ.ሜ፣ ሜ) ያረጋግጡ።
- የካርታ ጥያቄዎች፦ መጀመሪያ የአቅጣጫ ምልክቱን፣ ከዚያ ጎረቤቶችን ወይም ንብርብሮችን ይፈልጉ።

## Top 10 exam traps

### 10 የፈተና ወጥመዶች

| # | Trap | Correct |
|---|---|---|
| 1 | Who coined “geography”? | Eratosthenes (not Humboldt) |
| 2 | Largest country in Africa? | 10th largest (largest in the Horn) |
| 3 | Ethiopia has a Red Sea coast | Landlocked |
| 4 | Longest border | Somalia 1,600 km; shortest Djibouti 310 km |
| 5 | Higher A/B = weaker? | Higher A/B = more compact |
| 6 | Youngest Mesozoic rock | Upper sandstone (Cretaceous) |
| 7 | Rift lowest / narrowest | Afar lowest and widest; Lakes Region highest and narrowest |
| 8 | Ertale vs Fentale | Ertale active; Fentale dormant |
| 9 | Denudation = endogenic? | Exogenic |
| 10 | Sea direction in Triassic | SE → NW advance; retreat in Cretaceous |

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# Question Bank: 40 Practice Questions

## የመለማመጃ ጥያቄዎች ስብስብ፦ 40 ጥያቄዎች

### ENGLISH

Time: 45 minutes. Choose the best answer. 15 Easy · 15 Medium · 10 Hard/EUEE Trap. These are original questions written for this book (EUEE style), not copied past papers. Answers are in the Answer Key.

### አማርኛ

> ጊዜ፦ 45 ደቂቃ። ምርጡን መልስ ይምረጡ። 15 ቀላል · 15 መካከለኛ · 10 ከባድ/የፈተና ወጥመድ። እነዚህ ለዚህ መጽሐፍ የተጻፉ ኦሪጅናል ጥያቄዎች ናቸው። መልሶቹ በመልስ ቁልፉ ውስጥ አሉ።

**Q1 [Easy — Meaning] ጥያቄ 1**

**The word “geography” comes from the Greek words geo and graphos. What do they mean?**

- **A)** sea and map
- **B)** people and place
- **C)** earth and writing
- **D)** land and measure

**Q2 [Easy — Meaning] ጥያቄ 2**

**Who is credited with coining the term “geography”?**

- **A)** Eratosthenes
- **B)** Humboldt
- **C)** Hartshorne
- **D)** Yeates

**Q3 [Easy — Branches] ጥያቄ 3**

**Which branch of geography studies landforms and their distribution?**

- **A)** Climatology
- **B)** Biogeography
- **C)** Oceanography
- **D)** Geomorphology

**Q4 [Easy — Branches] ጥያቄ 4**

**Which of the following is a sub-branch of HUMAN geography?**

- **A)** Soil geography
- **B)** Urban geography
- **C)** Climatology
- **D)** Zoogeography

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### Q5 [Easy — Location] ጥያቄ 5

**How many countries share a border with Ethiopia?**

**A)** Four
**B)** Six
**C)** Five
**D)** Seven

### Q6 [Easy — Location] ጥያቄ 6

**With which neighbour does Ethiopia have its LONGEST boundary?**

**A)** Sudan
**B)** Kenya
**C)** Eritrea
**D)** Somalia

### Q7 [Easy — Location] ጥያቄ 7

**Ethiopia's absolute location in latitude is**

**A)** 3°N – 15°N
**B)** 3°S – 15°N
**C)** 33°N – 48°N
**D)** 0° – 12°N

### Q8 [Easy — Size] ጥያቄ 8

**The total area of Ethiopia is about**

**A)** 1,600,000 km²
**B)** 506,000 km²
**C)** 1,106,000 km²
**D)** 5,260 km²

### Q9 [Easy — Landforms] ጥያቄ 9

**The highest peak in Ethiopia is**

**A)** Tulu Dimtu
**B)** Batu
**C)** Gughe
**D)** Ras Dashen

### Q10 [Easy — Geological eras] ጥያቄ 10

**The oldest and longest geological era is the**

**A)** Paleozoic
**B)** Precambrian
**C)** Mesozoic
**D)** Cenozoic

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### Q11 [Easy — Geological eras] ጥያቄ 11

**Which era is known for the predominance of invertebrates?**

- **A)** Mesozoic
- **B)** Cenozoic
- **C)** Paleozoic
- **D)** Precambrian

### Q12 [Easy — Quaternary] ጥያቄ 12

**Which volcano in the Afar is ACTIVE?**

- **A)** Ertale
- **B)** Fentale
- **C)** Dallol
- **D)** Metahara

### Q13 [Easy — Landforms] ጥያቄ 13

**The lowest point in Ethiopia is the**

- **A)** Dallol depression
- **B)** Lake Turkana
- **C)** Awash valley
- **D)** Ogaden plain

### Q14 [Easy — Branches] ጥያቄ 14

**Zoogeography is the study of the distribution of**

- **A)** plants
- **B)** soils
- **C)** animals
- **D)** oceans

### Q15 [Easy — Mesozoic] ጥያቄ 15

**The oldest sedimentary rock in Ethiopia is**

- **A)** Hintalo limestone
- **B)** Adigrat sandstone
- **C)** Upper sandstone
- **D)** Trap basalt

### Q16 [Medium — Scope] ጥያቄ 16

**The part of the environment that has been modified by humans is the**

- **A)** lithosphere
- **B)** biosphere
- **C)** hydrosphere
- **D)** anthroposphere

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### Q17 [Medium — Location types] ጥያቄ 17

**“Ethiopia lies south-west of the Red Sea” describes its**

- **A)** absolute location
- **B)** strategic location
- **C)** vicinal location
- **D)** astronomical location

### Q18 [Medium — Shape] ጥያቄ 18

**If A = 1,106,000 km² and B = 5,260 km, the area–boundary ratio (A/B) is about**

- **A)** 210 km² per km
- **B)** 5.3 km² per km
- **C)** 1.41 km² per km
- **D)** 3,728 km² per km

### Q19 [Medium — Shape] ጥያቄ 19

**A B/C ratio closer to 1 shows that a country is**

- **A)** more elongated
- **B)** more fragmented
- **C)** less defensible
- **D)** more compact

### Q20 [Medium — Size] ጥያቄ 20

**In size, Ethiopia ranks in Africa as the**

- **A)** largest
- **B)** third largest
- **C)** tenth largest
- **D)** twentieth largest

### Q21 [Medium — Size] ጥያቄ 21

**Which is an ADVANTAGE of Ethiopia's large size?**

- **A)** Cheap administration
- **B)** Small army needed
- **C)** Diverse agro-ecological zones
- **D)** Easy social integration

### Q22 [Medium — Mesozoic] ጥያቄ 22

**Hintalo limestone was formed in the**

- **A)** Triassic period
- **B)** Cretaceous period
- **C)** Tertiary period
- **D)** Jurassic period

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### Q23 [Medium — Mesozoic] ጥያቄ 23

**During the Triassic, subsidence and sea advance proceeded from the**

- **A)** north-west towards the south-east
- **B)** south-east towards the north-west
- **C)** north towards the south
- **D)** east towards the west only

### Q24 [Medium — Rift Valley] ጥያቄ 24

**The Ethiopian Rift Valley was formed at the end of the**

- **A)** Tertiary period
- **B)** Precambrian era
- **C)** Triassic period
- **D)** Paleozoic era

### Q25 [Medium — Physiography] ጥያቄ 25

**Which physiographic division is the LARGEST (44 % of the country)?**

- **A)** The Western Highlands and associated lowlands
- **B)** The Southeastern Highlands
- **C)** The Rift Valley
- **D)** The Southeastern Lowlands

### Q26 [Medium — Physiography] ጥያቄ 26

**The main source region of the Wabe Shebelle and Genale rivers is the**

- **A)** Southwestern Highlands
- **B)** Southeastern Highlands
- **C)** Plateau of Tigray
- **D)** Afar Triangle

### Q27 [Medium — Physiography] ጥያቄ 27

**Which lowland of the west is the wettest because of consistent river flow?**

- **A)** Tekezze lowland
- **B)** Abbay–Dinder lowland
- **C)** Omo-Gibe lowland
- **D)** Baro-Akobo lowland

### Q28 [Medium — Rift Valley] ጥያቄ 28

**The smallest section of the Ethiopian Rift is the**

- **A)** Afar Triangle
- **B)** Main Ethiopian Rift
- **C)** Chew Bahir Rift
- **D)** Awash valley

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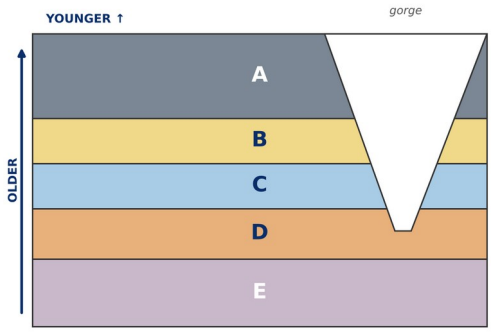
### Q29 [Medium — Landforms] ጥያቄ 29

**The relief (highest minus lowest altitude) of Ethiopia is**

**A)** 4,504 m
**B)** 4,620 m
**C)** 3,884 m
**D)** 4,736 m

### Q30 [Medium — Rock layers] ጥያቄ 30

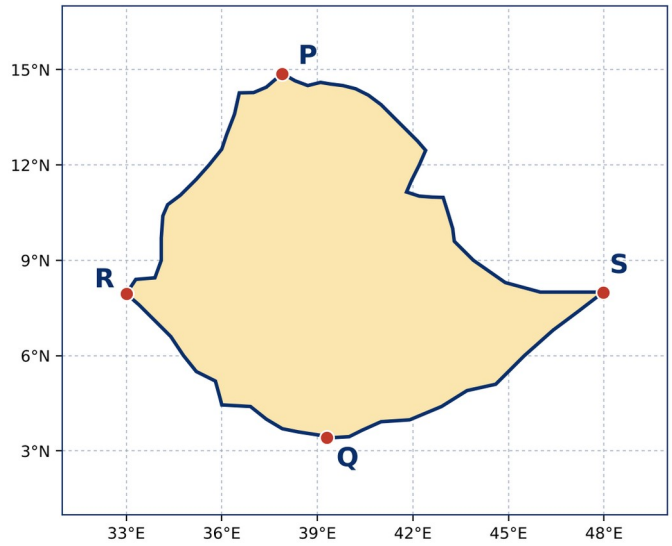
**In the figure, which layer is Hintalo limestone?**




**A)** B
**B)** D
**C)** C
**D)** E

### Q31 [Hard — Extreme points] ጥያቄ 31

**In the map, which point is Moyale (the southernmost point)?**




**A)** Q
**B)** P
**C)** R
**D)** S

### Q32 [Hard — Mesozoic] ጥያቄ 32

**Which pairing of rock and period is CORRECT?**

**A)** Adigrat sandstone – Cretaceous
**B)** Upper sandstone – Cretaceous (youngest sedimentary)
**C)** Hintalo limestone – Triassic
**D)** Trap basalt – Jurassic

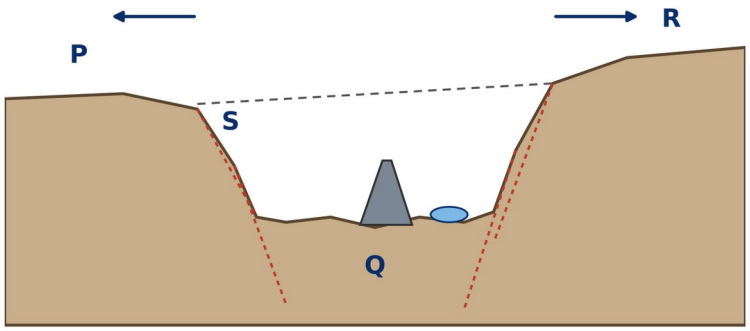
### Q33 [EUEE Trap — Rift Valley] ጥያቄ 33

**Which statement about the parts of the Rift Valley is INCORRECT?**

**A)** The Afar Triangle is the largest and widest part.
**B)** The Afar Triangle is the narrowest and most elevated part.
**C)** The Main Ethiopian Rift is the narrowest and most densely populated.
**D)** The Chew Bahir Rift is separated from the Lakes Region by the Konso Highlands.

### Q34 [Hard — Rift Valley] ጥያቄ 34

**In the rift cross-section, which letter marks the fault scarp (the steep edge of the valley)?**




**A)** P
**B)** Q
**C)** S
**D)** R

### Q35 [EUEE Trap — Landforms] ጥያቄ 35

**Which of these is NOT an endogenic force?**

**A)** Denudation
**B)** Volcanic activity
**C)** Tectonic movement
**D)** Faulting of the crust

### Q36 [Hard — Shape] ጥያቄ 36

**Country X has B/C = 1.25. Ethiopia has B/C = 1.411. Compared with Ethiopia, country X is**

**A)** less compact
**B)** equally compact
**C)** impossible to compare
**D)** more compact

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### Q37 [EUEE Trap — Physiography] ጥያቄ 37

**Which statement about the Plateau of Shewa is TRUE?**

**A)** It is the smallest, dome-shaped sub-division and a watershed of the Awash, Omo-Gibe and Abbay basins.
**B)** It is the largest sub-division and lies north of the Tekezze River.
**C)** It is a lowland drained only by the Wabe Shebelle.
**D)** It is the wettest region of Ethiopia with over 1,500 mm of rain.

### Q38 [Hard — Landforms] ጥያቄ 38

**If 56 % of Ethiopia (1,106,000 km²) lies above 1,000 m, the highland area is about**

**A)** 486,000 km²
**B)** 199,000 km²
**C)** 1,050,000 km²
**D)** 619,000 km²

### Q39 [EUEE Trap — Quaternary] ጥያቄ 39

**Which belongs to the QUATERNARY rather than the Tertiary?**

**A)** Trappean basalt flows
**B)** Uplift forming huge domes
**C)** Metahara lava fields and dormant Fentale
**D)** Formation of the Southeastern Highlands

### Q40 [EUEE Trap — Mesozoic] ጥያቄ 40

**Sandstone and limestone are seen at the surface in Tigray because**

**A)** they formed after the basalt
**B)** the overlying basalt was eroded away, exposing them
**C)** they were pushed up through the basalt by glaciers
**D)** the sea covers Tigray today

## Short-answer questions

አጫጭር መልስ ጥያቄዎች

**S1.** Define geography and name the scholar credited with coining the term. (3)

**S2.** Distinguish vicinal from strategic location, with one Ethiopian example each. (4)

**S3.** State two advantages and two disadvantages of Ethiopia's large size. (4)

**S4.** Calculate Ethiopia's A/B ratio and explain what it means for defence. (3)

**S5.** Describe the geological events of the Tertiary period in Ethiopia. (4)

**S6.** Compare the Afar Triangle and the Main Ethiopian Rift. (4)

# Answer Key with Explanations

## የመልስ ቁልፍ ከማብራሪያ ጋር

| Q1–10 | Q11–20 | Q21–30 | Q31–40 |
|---|---|---|---|
| **Q1: C** | Q11: C | Q21: C | Q31: A |
| **Q2: A** | Q12: A | Q22: D | Q32: B |
| **Q3: D** | Q13: A | Q23: B | Q33: B |
| **Q4: B** | Q14: C | Q24: A | Q34: C |
| **Q5: B** | Q15: B | Q25: A | Q35: A |
| **Q6: D** | Q16: D | Q26: B | Q36: D |
| **Q7: A** | Q17: B | Q27: D | Q37: A |
| **Q8: C** | Q18: A | Q28: C | Q38: D |
| **Q9: D** | Q19: D | Q29: D | Q39: C |
| **Q10: B** | Q20: C | Q30: C | Q40: B |

**Q1 [Easy — Meaning] Answer: C** መልስ

Geo = earth, graphos = writing: “writing about the earth”.

ጂኦ = ምድር፣ ግራፎስ = መጻፍ።

**Q2 [Easy — Meaning] Answer: A** መልስ

Eratosthenes (276–194 B.C.) coined it. The others gave later definitions.

ቃሉን የፈጠረው ኤራቶስቴንስ ነው፤ ሌሎቹ በኋላ ትርጓሜ ሰጥተዋል።

**Q3 [Easy — Branches] Answer: D** መልስ

Geomorphology = landforms.

ጂኦሞርፎሎጂ የመሬት ቅርጾችን ያጠናል።

**Q4 [Easy — Branches] Answer: B** መልስ

Urban geography studies the structure of cities; the others are physical.

የከተማ ጂኦግራፊ ሰዋዊ ነው፤ ሌሎቹ ተፈጥሯዊ ናቸው።

**Q5 [Easy — Location] Answer: B** መልስ

Djibouti, Eritrea, Kenya, Somalia, Sudan and South Sudan.

ጅቡቲ፣ ኤርትራ፣ ኬንያ፣ ሶማሊያ፣ ሱዳንና ደቡብ ሱዳን።

**Q6 [Easy — Location] Answer: D** መልስ

Somalia, 1,600 km (the shortest is Djibouti, 310 km).

ሶማሊያ፣ 1,600 ኪ.ሜ።

**Q7 [Easy — Location] Answer: A** መልስ

3°N–15°N latitude and 33°E–48°E longitude.

3°N–15°N ኬክሮስ።

**Q8 [Easy — Size] Answer: C** መልስ

1,106,000 km² (5,260 km is the boundary length).

1,106,000 ካ.ኪ.ሜ።

**Q9 [Easy — Landforms] Answer: D** መልስ

Ras Dashen, 4,620 m.

ራስ ዳሽን፣ 4,620 ሜ።

**Q10 [Easy — Geological eras] Answer: B** መልስ

Precambrian: 4.5 billion–600 million years ago.

ፕሪካምብሪያን።

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**Q11 [Easy — Geological eras] Answer: C** መልስ

Paleozoic = invertebrates; Mesozoic = reptiles.

ፓሊዮዞይክ።

**Q12 [Easy — Quaternary] Answer: A** መልስ

Ertale is active; Fentale is dormant.

ኤርታሌ ንቁ ነው።

**Q13 [Easy — Landforms] Answer: A** መልስ

Dallol, −116 m.

ዳሎል፣ −116 ሜ።

**Q14 [Easy — Branches] Answer: C** መልስ

Zoogeography = animals; phytogeography = plants.

ዞኦጂኦግራፊ እንስሳትን ያጠናል።

**Q15 [Easy — Mesozoic] Answer: B** መልስ

Adigrat sandstone formed in the Triassic.

የአድግራት አሸዋ ድንጋይ (ትራይያሲክ)።

**Q16 [Medium — Scope] Answer: D** መልስ

Anthroposphere = human-modified environment.

የሰው ስፌር።

**Q17 [Medium — Location types] Answer: B** መልስ

Position relative to a water body = strategic. Vicinal = neighbouring countries; absolute = lat/long.

ከውሃ አካል አንጻር = ስትራቴጂያዊ።

**Q18 [Medium — Shape] Answer: A** መልስ

1,106,000 ÷ 5,260 = 210.3

A/B = 1,106,000 ÷ 5,260 ≈ 210.

A/B ≈ 210።

**Q19 [Medium — Shape] Answer: D** መልስ

B/C = 1 for a perfect circle; closer to 1 means more compact.

ወደ 1 ሲቀርብ ይበልጥ ውሱን።

**Q20 [Medium — Size] Answer: C** መልስ

Tenth largest in Africa, largest in the Horn.

በአፍሪካ አሥረኛ።

**Q21 [Medium — Size] Answer: C** መልስ

Large size gives ecological variety; the other options are disadvantages in reverse.

ልዩ ልዩ የግብርና ሥነ ምህዳር።

**Q22 [Medium — Mesozoic] Answer: D** መልስ

Triassic → Adigrat; Jurassic → Hintalo; Cretaceous → Upper sandstone.

ጁራሲክ።

**Q23 [Medium — Mesozoic] Answer: B** መልስ

Subsidence began in the SE and the sea advanced NW.

ከደቡብ ምስራቅ ወደ ሰሜን ምዕራብ።

**Q24 [Medium — Rift Valley] Answer: A** መልስ

Formed at the end of the Tertiary by tectonic activity.

በተርሸሪ መጨረሻ።

**Q25 [Medium — Physiography] Answer: A** መልስ

Western Highlands = 44 %; the Rift = 18 %.

የምዕራብ ደጋ 44 %።

**Q26 [Medium — Physiography] Answer: B** መልስ

The Southeastern Highlands (Arsi, Bale, Hararghe, Sidama) feed these rivers.

የደቡብ ምስራቅ ደጋ።

**Q27 [Medium — Physiography] Answer: D** መልስ

The Baro river makes Baro-Akobo the wettest lowland.

ባሮ-አኮቦ።

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**Q28 [Medium — Rift Valley] Answer: C** መልስ

Chew Bahir (Omo–Gibe trough) is the smallest.

ጨው ባሕር ስምጥ።

**Q29 [Medium — Landforms] Answer: D** መልስ

4,620 − (−116) = 4,736

4,620 − (−116) = 4,736 m.

4,620 + 116 = 4,736።

**Q30 [Medium — Rock layers] Answer: C** መልስ

From the top: A trap basalt, B Upper sandstone (Cretaceous), C Hintalo limestone (Jurassic), D Adigrat sandstone (Triassic), E basement complex.

ከላይ፦ A ትራፕ ባሳልት፣ B የላይኛው አሸዋ፣ C ሂንታሎ ኖራ፣ D አድግራት፣ E የመሠረት ዐለት።

**Q31 [Hard — Extreme points] Answer: A** መልስ

P = north (Tigray), Q = south (Moyale), R = west (Akobo), S = east (Ogaden tip).

P ሰሜን (ትግራይ)፣ Q ደቡብ (ሞያሌ)፣ R ምዕራብ (አኮቦ)፣ S ምስራቅ (ኦጋዴን)።

**Q32 [Hard — Mesozoic] Answer: B** መልስ

Adigrat = Triassic; Hintalo = Jurassic; Upper sandstone = Cretaceous; trap basalt = Tertiary.

የላይኛው አሸዋ = ክሪቴሸስ።

**Q33 [EUEE Trap — Rift Valley] Answer: B** መልስ

The Afar Triangle is the widest (200–300 km) and LOWEST; the narrowest, highest part is the Main Ethiopian Rift.

የአፋር ሦስት ማዕዘን ሰፊውና ዝቅተኛው ነው፤ ጠባቡና ከፍተኛው ዋናው ስምጥ ነው።

**Q34 [Hard — Rift Valley] Answer: C** መልስ

P = Western Highlands plateau, Q = rift valley floor (graben), R = Southeastern Highlands, S = fault scarp.

P የምዕራብ ደጋ፣ Q የስምጥ ወለል፣ R የደቡብ ምስራቅ ደጋ፣ S የጥፋት ግድግዳ።

**Q35 [EUEE Trap — Landforms] Answer: A** መልስ

Denudation (erosion) acts at the surface: exogenic.

መሸርሸር የውጭ (exogenic) ኃይል ነው።

**Q36 [Hard — Shape] Answer: D** መልስ

1.25 is closer to 1 than 1.411, so X is more compact.

1.25 ወደ 1 ይቀርባል፤ ስለዚህ X ይበልጥ ውሱን።

**Q37 [EUEE Trap — Physiography] Answer: A** መልስ

The Shewa plateau is the smallest, dome-shaped western sub-division and a watershed. The wettest region is the Southwestern Highlands.

የሸዋ አምባ ትንሹ የጉልላት ቅርጽ ያለው የተፋሰስ ክፍፍል ነው፤ እርጥብ የደቡብ ምዕራብ ደጋ ነው።

**Q38 [Hard — Landforms]** **Answer: D** መልስ

$0.56 \times 1{,}106{,}000 = 619{,}360\ km^2$

$0.56 \times 1{,}106{,}000 = 619{,}360\ km^2$.

$0.56 \times 1{,}106{,}000 = 619{,}360$።

**Q39 [EUEE Trap — Quaternary]** **Answer: C** መልስ

Tertiary: uplift, trap lava, highlands, rift formation. Quaternary: recent volcanism (Ertale, Fentale, Metahara), hot springs.

ተርሸሪ፦ ከፍታ፣ ትራፕ ላቫ፣ ደጋዎች፤ ኳተርነሪ፦ የቅርብ ጊዜ እሳተ ገሞራ።

**Q40 [EUEE Trap — Mesozoic]** **Answer: B** መልስ

The Mesozoic sedimentary rocks were buried by Cenozoic basalt, which erosion later removed in places.

ባሳልቱ ተሸርሽሮ ስለተወገደ።

## Short-answer model answers

የአጫጭር ጥያቄዎች ናሙና መልሶች

**S1.** Define geography and name the scholar credited with coining the term. (3)

Geography is the scientific study of the Earth that describes and analyses spatial and temporal variations of physical, biological and human phenomena and their interrelationships. Eratosthenes (276–194 B.C.) coined the term.

ጂኦግራፊ የምድርን ተፈጥሯዊ፣ ሕይወታዊና ሰዋዊ ክስተቶች የቦታና የጊዜ ልዩነቶችና ግንኙነቶች የሚተነትን ሳይንሳዊ ጥናት ነው። ቃሉን የፈጠረው ኤራቶስቴንስ ነው።

**S2.** Distinguish vicinal from strategic location, with one Ethiopian example each. (4)

Vicinal location is position relative to neighbouring countries (e.g. Ethiopia borders Somalia and Kenya). Strategic location is position relative to major water bodies and landmasses (e.g. south-west of the Red Sea, within the Nile Basin).

አጎራባች አቀማመጥ ከጎረቤት አገሮች አንጻር ያለ ቦታ ነው (ለምሳሌ ከሶማሊያና ኬንያ ጋር ትዋሰናለች)።
ስትራቴጂያዊ አቀማመጥ ከውሃ አካላትና ከየብስ አንጻር ያለ ቦታ ነው (ለምሳሌ ከቀይ ባሕር ደቡብ ምዕራብ፣ በአባይ ተፋሰስ ውስጥ)።

**S3.** State two advantages and two disadvantages of Ethiopia's large size. (4)

Advantages: diverse agro-ecological zones/fauna and flora; large arable land; variety of minerals; diverse peoples. Disadvantages: costly administration; large army needed; heavy infrastructure costs; difficult social and economic integration. (any two of each)

ጥቅም፦ ልዩ ልዩ ሥነ ምህዳር፣ ሰፊ የሚታረስ መሬት፣ የማዕድን ዓይነቶች፣ የተለያዩ ብሔረሰቦች። ጉዳት፦ ውድ አስተዳደር፣ ትልቅ ሠራዊት፣ የመሠረተ ልማት ወጪ፣ ውህደት አስቸጋሪነት። (ከእያንዳንዱ ሁለት)

**S4.** Calculate Ethiopia's A/B ratio and explain what it means for defence. (3)

A/B = 1,106,000 ÷ 5,260 ≈ 210 $km^2$ per km. Defending 1 km of border protects about 210 $km^2$ of

hinterland, which favours defence.
A/B = 1,106,000 ÷ 5,260 ≈ 210 ካ.ኪ.ሜ በኪ.ሜ። አንድ ኪ.ሜ ድንበር መጠበቅ ወደ 210 ካ.ኪ.ሜ የውስጥ መሬትን ይጠበቃል፤ ይህም መከላከልን ያግዛል።

**S5.** Describe the geological events of the Tertiary period in Ethiopia. (4)
Uplift reached a maximum forming dome-like structures; great cracks opened and basaltic (Trappean) lava flowed; the Northwestern and Southeastern Highlands and Somali plateaus formed; cracking and faulting created the Great East African Rift Valley System.
ከፍታ ጉልላት መሰል መዋቅሮችን ፈጠረ፤ ታላላቅ ስንጥቆች ተከፈቱ፤ የባሳልት (ትራፕ) ላቫ ፈሰሰ፤ የሰሜን ምዕራብና የደቡብ ምስራቅ ደጋዎችና የሶማሊ አምባዎች ተፈጠሩ፤ ስንጥቅና ጥፋት የምስራቅ አፍሪካ ስምጥ ሥርዓትን ፈጠሩ።

**S6.** Compare the Afar Triangle and the Main Ethiopian Rift. (4)
Afar Triangle: northern, largest and widest (200–300 km), lowest (−116 m to 900 m), salt plains, grabens, volcanoes, fossil sites. Main Ethiopian Rift (Lakes Region): central, most elevated and narrowest, wettest and most densely populated, sedentary farming, many lakes.
የአፋር ሦስት ማዕዘን፦ ሰሜናዊ፣ ትልቁና ሰፊው (200–300 ኪ.ሜ)፣ ዝቅተኛ፣ የጨው ሜዳዎች፣ እሳተ ገሞራዎች፣ ቀረተ አካል ቦታዎች። ዋናው ስምጥ (የሐይቆች አካባቢ)፦ መካከለኛ፣ ከፍተኛውና ጠባቡ፣ እርጥብና ብዙ ሕዝብ ያለበት፣ ቋሚ እርሻ፣ ብዙ ሐይቆች።

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# Coverage Verification

## የይዘት ሽፋን ማረጋገጫ

### ENGLISH

Each topic in the source notes is mapped to the section of this book that teaches it and the questions that test it.

አማርኛ

> በምንጭ ማስታወሻው ውስጥ ያለ እያንዳንዱ ርዕስ በዚህ መጽሐፍ ከሚያስተምረው ክፍልና ከሚፈትኑት ጥያቄዎች ጋር ተያይዟል።

| Source topic | Section | Worked example | Questions |
|---|---|---|---|
| **1.1.1 Meaning, Eratosthenes, definitions** | 1.1.1 | 1.1 | Q1, Q2, S1 |
| **1.1.2 Scope, five spheres** | 1.1.2 | 1.2 | Q16 |
| **1.1.3 Branches (physical, human)** | 1.1.3 | 1.3 | Q3, Q4, Q14 |
| **1.2.1 Vicinal, strategic, absolute location** | 1.2.1 | 1.4–1.5 | Q5–Q7, Q17, Q31, S2 |
| **1.2.2 Size, advantages, disadvantages** | 1.2.2 | 1.6 | Q8, Q20, Q21, S3 |
| **1.2.3 Shape, B/C, A/B** | 1.2.3 | 1.7–1.9 | Q18, Q19, Q36, S4 |
| **1.3.1 Geological eras (Precambrian–Cenozoic)** | 1.3.1 | 1.10–1.12 | Q10, Q11, Q15, Q22–Q24, Q30, Q32, Q39, Q40, S5 |
| **Quaternary volcanism (Ertale, Fentale, Metahara)** | 1.3.1 | 1.12 | Q12, Q39 |
| **1.3.2 Altitude extremes, highland share, forces** | 1.3.2 | 1.13 | Q9, Q13, Q29, Q35, Q38 |
| **Western Highlands and Lowlands** | 1.3.2 | 1.15 | Q25, Q27, Q37 |
| **Southeastern Highlands and Lowlands** | 1.3.2 | 1.15 | Q26 |
| **Rift Valley: Afar, Main Rift, Chew Bahir** | 1.3.2 | 1.14–1.15 | Q24, Q28, Q33, Q34, S6 |

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**About 8.5 / 10.** All source topics are covered with bilingual explanations, 15 worked-example boxes, misconception and takeaway boxes, and 40 questions (15 Easy, 15 Medium, 10 Hard/Trap) with explanations. Limits: (1) the maps are schematic hand-drawn outlines, labelled as such, with textbook values on the labels; (2) the questions are original, not verified against real EUEE papers; (3) the Amharic should be read once by a subject teacher for terminology preferences; (4) the source's own Review Exercise had answers without question stems, so new questions were written instead.

ወደ **8.5 / 10**። ሁሉም የምንጭ ርዕሶች በሁለት ቋንቋ ተሸፍነዋል። ገደቦች፦ (1) ካርታዎቹ ንድፍ ናቸው፤ (2)

ጥያቄዎቹ ኦሪጅናል ናቸው እንጂ ከእውነተኛ EUEE ፈተናዎች ጋር አልተረጋገጡም፤ (3) አማርኛውን የትምህርቱ መምህር አንድ ጊዜ ቢያነበው ይመከራል።

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